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BUBBLEVERSE

Technical Appendices

Building a Model of the Universe from Observation

Appendices A–D

Computational methods and provenance

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Appendix A — Computational Provenance Ledger

Historical computational provenance — superseded active interpretation. The following attribution remains valid for the original frozen parameter points that were evaluated, but its $n=3$ EDE TE/intermediate- ℓ pattern must not be treated as the active description of the corrected high- H_0 basins. Later multi-start minimization and corrected fixed-point attribution supersede that interpretation while preserving the original calculation as computational history. The subsequent fixed- H_0 likelihood-attribution analysis used the same frozen common backend to profile $n=3$ early dark energy and varying electron mass at their respective reference points and at $H_0 = 71.5, 72.5$, and $73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$. It decomposed the resulting fit changes into non-overlapping Planck low- ℓ , ACT DR6, DESI DR2, and BBN contributions while profiling the remaining permitted parameters and nuisance quantities. All required job-completeness, component-attribution, and final-result gates passed. The calculation should be interpreted as a common-backend likelihood diagnostic rather than as an exact reproduction of the K-036 or K-038 native analyses. Global minima have not been established for every fixed- H_0 profile, and later globality and fixed-point attribution results supersede earlier high- H_0 $n=3$ EDE profile points where lower basins were subsequently found.

The exact GitHub run ID, execution commit, result identifier, and result-file hash for this completed fixed- H_0 likelihood-attribution run are NOT DOCUMENTED IN THE WORD FILE. A subsequent observable-level ACT DR6 attribution evaluated the same eight frozen profile points—four $n=3$ EDE points and four varying-electron-mass points—without performing a new optimization. The calculation reconstructed the ACT DR6-lite quadratic form using the full bandpower covariance and assigned signed per-bandpower contributions according to

$C_i = d_i (C_{-1d})_i$.

T C

The sum of these contributions was required to reproduce d_{-1d} , allowing TT, TE, EE, and

multipole-region attributions to be reported without falsely treating them as statistically independent likelihoods. All eight required points completed, and the covariance-interpretation, job-completeness, quadratic-closure, and final-result gates passed.

For $n=3$ EDE, TE was the largest positive spectrum-level contribution at each tested high-

H_0 point. For varying electron mass, the ACT response remained approximately neutral at $H_0=72.5$ and $73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ because positive and negative covariance-weighted

spectral contributions substantially cancelled.

The attribution used ACTCollaboration/DR6-ACT-lite commit

0e0cd2c703c62a0e980470b572602233b27750e1 and the frozen common-backend chain

based on class_edc commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97. The internal

result identifier recorded by the aggregate is Q007-ACT-OBSERVABLE-ATTRIBUTION-V1; the

corrected execution was run through the "Bubbleverse Q007 ACT Observable Attribution V2" workflow. The exact successful GitHub run ID and head commit are not preserved in the supplied final aggregate and therefore remain [PROVENANCE TO BE VERIFIED].

The observable attribution does not establish global minima for the underlying fixed-H 0 profiles. It should be read as a technically validated decomposition of the specified frozen points. If later minimizer or globality tests materially change those profile minima, the present attribution remains valid for the original points but would need to be superseded as the active quantitative description.

A subsequent, broader multi-start optimization materially sharpened this conclusion.

Using the same frozen scientific configuration and likelihood surface, the calculation again
-1 M pc

examined fixed H 0=70.853292,71.5,72.5, and 73.5k m s

-1, but with a substantially

expanded set of independent starting conditions and refinement searches. The resulting combined classification remained BEST-OBSERVED MINIMUM ONLY: none of the tested points achieved a mathematical proof of globality, and the predefined criterion for strong numerical near-global certification was not satisfied. The expanded search therefore strengthens the evidence that optimizer-basin structure is scientifically relevant while explicitly failing to establish that the currently observed minima are the final global profile.

-1 M pc

The most important revision occurs at H 0=71.5k m s

-1. The broader search finds

2=546.364260, compared with the previous best-observed value of 547.154321. The

χ

improvement, $\Delta \chi$

2=-0.790061, exceeds the predefined material-improvement threshold of 0.50, so the earlier H 0=71.5 minimum is superseded as the active best-observed solution.

At H 0=72.5, the newer search reaches χ

2=548.624632, an improvement of 0.230023, while

at H 0=73.5 it reaches χ

2=548.766646, an improvement of 0.338938. These latter changes do

not cross the predefined material-improvement threshold, although the lower values

remain relevant as newer best-observed candidates. The same search also reaches

χ

2=546.679939 at H 0=70.853292, compared with 548.321909 in the earlier validation search, further demonstrating that the finite-search likelihood envelope is sensitive to optimizer-basin discovery. [Internal computational provenance: INTERNAL-Q005,

INTERNAL-Q008, R-Q011-EDE-GLOBALITY-CERTIFICATION-004.]

The validation found substantially lower high-H 0 solutions than the original fixed-profile

2=547.154321 at H 0=71.5, 548.854655 at 72.5, and 549.105584 at 73.5k m s

calculation: χ

-1 M pc

-1. At H 0=70.853292, the validation run itself obtained χ

2=548.321909, but

an earlier common-backend calculation had already documented the lower basin

χ

2=548.010401. The active best-observed envelope therefore retains the lowest

documented solution from either calculation rather than replacing the earlier basin with a

worse one.

The earlier validation values χ

$\chi^2=547.154321, 548.854655$, and 549.105584 at $H_0=71.5, 72.5$,

-1 M pc

and 73.5 k m s

-1 , respectively, remain important computational provenance, but they

no longer constitute the complete numerical picture. In particular, the $H_0=71.5$ point has

been materially superseded by the lower χ

$\chi^2=546.364260$ basin found in the expanded

search. The smaller subsequent improvements at $H_0=72.5$ and 73.5 do not satisfy the predefined 0.50 material-improvement gate. The active interpretation is therefore not that a final high- H_0 profile has been established, but that the best-observed envelope remains sensitive to basin discovery and must be treated as numerically provisional.

Computational provenance — expanded EDE globality certification. The subsequent robustness calculation is recorded as R-Q011-EDE-GLOBALITY-CERTIFICATION-004 and was executed through the “Bubbleverse Q011 EDE Globality Certification V4” workflow in the Morfindien/Bubbleverse repository. The execution used repository commit `e113fb846948f424b865d1303fba8b5988ec9133` and GitHub run `33363067156`. The numerical program was `q011_edg_globality_certification_v4.py`, SHA-256 `53dc99d00d220671fe16d17136ef626a7db454711120f5861a6677b3879c57e6`, with configuration `q011_edg_globality_certification_v4_config.yml`, SHA-256 `37baae48fd4f2bf70807dd2c6baab6e8a138725f8b70987773d4ddad5b8467b`. The run used Python 3.10.21, class_edg commit `5a131c91d657dd9a7c6364cc45b038710f8d0d97`, ACT DR6-lite commit `0e0cd2c703c62a0e980470b572602233b27750e1`, and the frozen Q005 V14 scientific configuration. It is an internal robustness calculation on the same underlying likelihood surface and must not be counted as independent observational evidence for early dark energy.

Neither the earlier validation nor the expanded search establishes mathematical global minima. More importantly, the expanded calculation fails to achieve the predefined strong numerical near-global certification across the tested high- H_0 targets. Its combined classification is therefore BEST-OBSERVED MINIMUM ONLY. The material improvement at $H_0=71.5$, together with the lower basin recovered at the reference target, strengthens the conclusion that optimizer-basin sensitivity is not a negligible technical detail: it can materially change the apparent high- H_0 likelihood profile. Finite multi-start optimization nevertheless cannot determine whether still lower basins remain undiscovered.

Computational provenance for this validation: INTERNAL-Q008, result class EDE_FIXED_H0_GLOBALITY_VALIDATION, run identifier Q008-EDE-GLOBALITY-VALIDATION-V1. [PROVENANCE TO BE VERIFIED: insert the successful GitHub workflow run ID, execution commit, and final result-file SHA-256 if these are to be retained in the published reproducibility record.

Computational provenance — corrected ACT DR6 attribution for $n=3$ early dark energy. The corrected fixed-point observable-attribution calculation is recorded as INTERNAL-Q009, run Q009-ACT-CORRECTED-MINIMA-ATTRIBUTION-V8, using `q009_act_corrected_minima_attribution_v8.py` with `q009_act_corrected_minima_attribution_v8_config.yml` in the “Bubbleverse Q009 ACT Corrected Minima Attribution V8” workflow. The repository is Morfindien/Bubbleverse; the repository inspection state used for the analysis was commit `a8d11cc9687e5df757f4794224d83ed34df03846`. The ACT DR6-lite likelihood was frozen at commit `0e0cd2c703c62a0e980470b572602233b27750e1`. Code version 8.0-q007-fixed-vector-consumption-fixed evaluated the complete selected best-fit vectors at fixed H_0 without reoptimization, sampling, new priors, new bounds, or additional datasets. This analysis is

dependent on the INTERNAL-Q005, INTERNAL-Q007, and INTERNAL-Q008 computational chain and must not be counted as independent observational evidence. Its conclusions apply to the tested best-observed basins; mathematical globality remains unresolved.

Computational provenance — Hubble-tension model comparison. The common-backend numerical analysis used mwt5345/class_edc commit

5a131c91d657dd9a7c6364cc45b038710f8d0d97 and ACTCollaboration/DR6-ACT-lite commit

0e0cd2c703c62a0e980470b572602233b27750e1, with Python 3.10.x, Cobaya 3.5.6, NumPy

1.26.4, SciPy 1.15.3, Py-BOBYQA 1.5.0, Cython 0.29.37, and SACC 1.0.2. Repeated common-

backend runs used GitHub run Ids 33249784822 and 33252632548 at commit

b875ea95d9b6390b8f03dac729eb7f5fe84d3118. The deterministic globality-recovery

calculation used GitHub run ID 33271326245 and workflow commit

7293c7f5db4d7ab72b1aaad38fe15f2639e11d5e; the corresponding internal result is R-Q005-

V14-GLOBALITY-RECOVERY-001, with result-JSON SHA-256

3642aa6e4d09c595610875cfed321f3d28055fb059757c2381f08007bf071253. These

calculations provide a controlled common-backend comparison and numerical basin

diagnostic; they are not paper-native reproductions of the Poulin et al. Or Toda and Seto implementations.

Computational provenance — covariance-aware Planck/SPT attribution for the $n=3$ EDE

high- H_0 region. The accepted attribution is recorded as R-Q015-EDE-CMB-

ATTRIBUTION-003, run Q015-CMB-ATTRIBUTION-V3. The scientific endpoint evaluations

were produced by the preceding Q015 V2 workflow, GitHub Actions run 33629190153 at

repository head 78a65aae897fca18132a5c0758bccdbbc6a09402. Its frozen-parent Stage-0

artifact was 9846055938 with SHA-256 digest

7adb0624a73f8071f2b4dfcac6e65ebfe2d4076277156a87b090c0cab7f99696; the Planck

endpoint artifact was 9846143033 with SHA-256 digest

ff6439352ae00ad0c87befb07b035342abe5d1fecf5dee09b4eb84f241bad3a6; and the SPT

endpoint artifact was 9846127249 with SHA-256 digest

b0740d6424b7f1fb85a6adf92a9bbd2205b27479fec2e7d2f6d97ee8c2be0c33. All three

scientific jobs completed successfully.

Computational provenance — matched Planck/SPT/ACT $n=3$ EDE response. The subsequent

matched-surface analysis is recorded as R-Q016-EDE-MATCHED-CMB-SURFACE-016. It used

the frozen mwt5345/class_edc backend at commit

5a131c91d657dd9a7c6364cc45b038710f8d0d97 and compared Planck, SPT-3G, and ACT

using TT, TE, and EE over $600 \leq \ell \leq 2000$, with each likelihood retaining its own nuisance

structure. Absolute likelihood values were not summed or compared across experiments.

–1 Mpc

The analysis produced separately certified free- H_0 and fixed- $H_0=71.5$ km s

–1

endpoints and reproduced the low- H_0 Planck basin under the predefined multistart

criteria.

The covariance-aware matched residual analysis is recorded as R-Q016-EDE-CMB-

MECHANISM-004, run Q016-CMB-MECHANISM-V4. The successful GitHub Actions

execution was run 33836015653 at repository head

91bc0736a00e2454d4e37324e5d3a1a7fe7813d6. Its FINAL_RESULT_GATE passed. Signed residual contributions were evaluated using the full within-experiment covariance and grouped into TT, TE, and EE and the common multipole intervals 600!–!999, 1000!–!1499, and 1500!–!2000. These allocations are covariance-coupled and are not independent χ measurements. The calculation found no common non-zero residual-shift sign across all three experiments in any of the nine predefined observable-by-multipole cells.

The matched analysis reuses the underlying Planck, SPT-3G, and ACT observational data and therefore constitutes an internal comparative likelihood analysis rather than independent observational evidence for early dark energy. Shared cosmic variance was not represented by a unified cross-experiment covariance, so no combined Planck–SPT–ACT significance is assigned. The attempted secondary full-multifrequency SPT and ACT mechanism stages remained technically unavailable; these implementation limitations do not invalidate the certified matched primary results.

The V2 final merge failed only because its validation logic required an effectively exact identity between the raw covariance quadratic used for bandpower attribution and the serialized likelihood component. The subsequent V3 certification changed that validation semantics without changing the scientific surface, model, endpoints, likelihoods, covariance matrices, theory calculation, optimizer results, or signed attribution values. The residual bridge terms were -0.00286835 for Planck and -0.00567713 for SPT and were retained explicitly rather than redistributed among spectra or multipole bins. The resulting driver rankings and scientific interpretation passed the V3 mandatory validation gates. V3 therefore constitutes a certification and merge of the already completed endpoint calculations rather than a new theory or optimization run.

The supplied final Q015 V3 archive has SHA-256 digest

80b774f66c9f0650e2c36f19633a2daa6fbee69d1dbde2a9df91e286520ca8d4. [PROVENANCE TO BE VERIFIED: insert the final Q015 V3 GitHub Actions run ID if it is to be retained in the published reproducibility record.] As with the preceding Bubbleverse calculations, this is an

analysis of existing observational likelihoods and must not be counted as an independent observational detection of early dark energy.

Computational provenance — full-multifrequency Planck nuisance-direction localization. The full-multifrequency Planck analysis is recorded as R-Q017-EDE-PLANCK-DIRECTION-LOCALIZATION-003, run Q017-PLANCK-DIRECTION-LOCALIZATION-V3. The successful GitHub Actions execution was run 33842355384 at repository commit

277bb274e5783831303b542681e8cf87a70d31c2. The calculation retained the frozen $n=3$ backend mwt5345/class_edex at commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97 and used the public planck_NPIPE_highl_CamSpec.TTTEEE likelihood. The certified cosmological endpoints from R-Q016-EDE-MATCHED-CMB-SURFACE-016 were held fixed rather than reoptimized. Nine Planck calibration and foreground nuisance parameters were reprofiled, using three independent starts at each free/fixed baseline endpoint and two starts for each nuisance-lock group. The optimizer configuration used BOBYQA with

priors retained, max_evals = 1280, and rhoend = 0.002. Baseline stability, lock-profile stability, endpoint/decomposition closure, the nested-lock non-negativity test, and the final-result gate all passed. The authoritative final classification is INDETERMINATE between a specifically localized and a broadly distributed Planck penalty. The final GitHub artifact is q017-final-v3, artifact ID 9925585139, with SHA-256 digest a45ba197f20595013d390136e00f8fd83f40a70fb9e9e1d83f2048f2d70320db. The calculation is an internal reanalysis of existing Planck observations and must not be counted as an independent observational detection or as evidence that a particular Planck systematic has been causally identified.

Computational provenance — Planck preferred-region and cross-profile geometry. Independent reprofiling of the frozen n=3 early-dark-energy model on the matched CamSpec-lite and full-multifrequency CamSpec likelihoods established that similar projected minima in H 0 and f E D E do not imply equivalent multidimensional preferred regions. The preserved historical cross-profile costs are +0.8120365009 when the full-multifrequency preferred cosmology is evaluated on CamSpec-lite and +13.5564057804 for the reverse evaluation. The authoritative preferred-region result is R-Q019-EDE-PLANCK-COSMOLOGY-REPROFILE-003. The earlier R-Q019-EDE-PLANCK-COSMOLOGY-REPROFILE-001 interpretation is superseded, but its valid endpoint and cross-profile calculations remain part of the numerical provenance.

The subsequent block-localization analysis is recorded as R-Q020-EDE-PLANCK-CROSS-PROFILE-DIRECTION-002, run Q020-PLANCK-CROSS-PROFILE-DIRECTION-REPAIR-V2. The successful GitHub Actions execution was run 33878880397 at repository commit b0c0a8fd89408bac91502e15d42c69820e913ab6. The final artifact q020-final-v2 has SHA-256 digest 9fed5d755abdb3630bbe3a1a88e23b0c8172b49dab129a4421b527c3bdabde7f. The calculation retained the frozen mwt5345/class_edc backend at commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.

Computational provenance — internal primordial-block structure. The primordial block identified by the preceding four-block analysis was subsequently decomposed into ns, log A , and τ_{reio} using the same frozen n=3 early-dark-energy backend, mwt5345/class_edc at commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97. The authoritative result is R-Q021-EDE-PRIMORDIAL-INTERNAL-STRUCTURE-002, run Q021-PRIMORDIAL-INTERNAL-**STRUCTURE-REPAIR-V2**.

The underlying numerical calculation comprised 48 finite primordial-profile evaluations: two likelihood constructions, eight primordial vertices, and three optimizer restarts. The original numerical execution, R-Q021-EDE-PRIMORDIAL-INTERNAL-STRUCTURE-001, remains the source of these raw profiles, but its final aggregation is superseded. That aggregation selected the lowest objective independently at each vertex and could therefore combine vertices from different optimizer restarts into a decomposition corresponding to no single coherent optimization trajectory. Its likelihood-identity validation also serialized live Python function representations containing process-specific memory addresses. Neither defect invalidated the 48 underlying likelihood evaluations.

The repaired analysis introduced no new likelihood evaluations. Instead, it constructed one complete decomposition for each optimizer restart, replaced the process-dependent likelihood hash with a stable structural identity test, preserved the observed optimizer-basin spread, and prohibited best-per-vertex mixing. The Q-identity, parent-identity, parent-reproduction, completeness, finite-result, structural-likelihood-identity, coherent-restart, decomposition-closure, structure-stability, no-Frankenstein-mixing, no-cross-chain- χ

2-sum, interpretation-safety, and final-result gates passed.

All three coherent full-multifrequency decompositions and all three coherent matched CamSpec-lite decompositions are classified as distributed. The result therefore supports a robust qualitative conclusion about the internal primordial structure while retaining uncertainty in its detailed quantitative attribution. The calculation is an internal reanalysis of existing Planck likelihood information and must not be counted as an independent observational detection, a causal identification of a Planck systematic, or evidence for new physics. The relevant external methodological and scientific context is provided by Efstathiou, Rosenberg & Poulin, Physical Review Letters 132, 221002 (2024), arXiv:2311.00524; Rosenberg, Gratton & Efstathiou, Monthly Notices of the Royal Astronomical Society 517, 4620 (2022), arXiv:2205.10869; and the Cobaya 3.5.6 technical documentation.

The scientific grid consisted of 64 complete and finite hybrid-vertex nuisance-profile evaluations inherited from the preceding Q020 V1 calculation. The final V2 validation corrected a denominator mismatch in the original validation gate rather than rerunning or altering those likelihood evaluations. It explicitly preserved the historical Q019 values, reproduced the absolute Q019 foreign-cosmology cross-profile objectives, required non-regression of the newly reprofiled native endpoints, and retained the prohibition against treating correlated likelihood constructions as additive physical χ

2 components. The

FINAL_RESULT_GATE passed, yielding the classification LIMITED MULTIDIMENSIONAL **DIRECTION LOCALIZED.**

Within the predefined four-block geometry, the full-multifrequency response assigns an absolute Shapley fraction of 0.7157833 to the primordial block (ns,log A , τ r eio), 0.1727489 to EDE timing, 0.0914685 to matter, and 0.0199994 to H 0. These allocations and the associated pair interactions are non-independent, non-causal functional-geometry diagnostics. They do not constitute a physical decomposition of the Planck likelihood and do not identify a particular instrumental systematic or new physical mechanism. The calculation is an internal reanalysis of existing likelihood information rather than independent observational evidence.

Four controlled full-multifrequency surfaces were evaluated: the native construction, the same likelihood with prior-density contributions removed from the minimization objective, a surface profiling only A pl anc k, c alT E, and c al E E while fixing the six multifrequency foreground parameters, and the reciprocal surface profiling the six foreground parameters while fixing those three shared nuisance parameters. Three independent starts were used for each bridge-and-endpoint combination, giving 24 atomic

profile calculations. The resulting fixed-minus-free penalties were 17.3953724212, 17.1601717140, 11.7641347631, and 15.6176804336, respectively.

The native-reference reproduction, endpoint identity, model identity, job completeness, multistart replication, finite-result, and non-additivity validation gates all passed. The reduced shared-nuisance surface exhibited substantial multistart spread and is therefore retained only as an unstable diagnostic. The authoritative scientific classification is COUPLED COMPONENTS / LIKELIHOOD-ARCHITECTURE DEPENDENCE: the difference between the matched and full-multifrequency Planck penalties cannot be assigned to one independent additive likelihood component. This calculation is an internal reanalysis of existing Planck observations and must not be counted as an independent observational detection, a causal identification of a Planck systematic, or evidence for new physics. Computational provenance — full-multifrequency basin geometry and exact interaction decomposition. The authoritative optimizer-basin continuation is R-Q022-EDE-FULLMF-OPTIMIZER-BASIN-002 at execution commit

16a7902c3be7d1967adaae9ee380baf0d17e903a. The subsequent fixed-lineage direction analysis R-Q023-EDE-FULLMF-BASIN-DIRECTIONS-002 was executed in GitHub Actions run 33913077353 at commit ea5af9a0b25090eb8472adb3c9574dcb4e1efe5c. The permutation- and sign-invariant matching analysis R-Q024-EDE-FULLMF-BASIN-MATCHING-001 used GitHub Actions run 33915600167 at commit b8e16629bbc6592ee11f0d90f78c887a913ac4c1. The fixed-vector component attribution R-Q025-EDE-FULLMF-BASIN-COMPONENT-ATTRIBUTION-002 used run 33918779905 at commit

b2db6827b1caf5405e6a34bff221b6ae4813c1d0, and the coordinate-level localization R-

Q026-EDE-FULLMF-MASK6-COUPLING-001 used run 33921255497 at commit

7cb4d98c01196c22b407b6f4fd4118179afea3de. The residual-and-covariance bridge R-Q027-EDE-FULLMF-RESIDUAL-COVARIANCE-BRIDGE-001 used run 33924573651 at commit df03e30712690e61ff799558598405a22c0bacbb.

The validated common-support calculation R-Q028-EDE-FULLMF-COMMON-SUPPORT-COUNTERFACTUAL-008 used GitHub Actions run 33957937309 at execution-head commit 7f5d35c0725530039162adfa4d71be288e8b4462. Its exact labelled common support contains 9915 of 9915 dimensions, with authoritative key SHA-256

6088165823b7fae224ce51aaef33fdef5a400aefafd1675e454e00c6c1a25a63. The final artifact q028-final-v8 has artifact ID 9967068363 and SHA-256 digest

7dfbd1f27251ff311dd1de6118539dd691ab03d9afdb08539b3b13586c50d5af. The previously recorded execution SHA 7f5d35efeb54b4a62d6dfb8f7074014b4fabcb190 and common-support hash 054220c018e58e902eab98b879eec4a823fb3238a1774ddbacc649bddca8b666 are superseded provenance misrecords and must not be used as active identifiers.

The exact residual-family identity subsequently tested numerically is

T P Δ

T P m).

2(m

F=−2d

2m+Δ

Its authoritative final numerical validation is R-Q030-ORIGINAL-GATE-PRECISION-AUDIT-001. The finalization workflow .github/workflows/q030-original-gate-audit-finalization-r1.yml completed successfully in GitHub Actions run 33963221940 at head SHA

4cbe1585d2603b507679c654a8ff8b0d7e2e522e. The final artifact q030-original-gate-precision-audit-final-v1 has artifact ID 9968592047 and SHA-256 digest 7685d7693f2e85391767bd9a2654a940280a02c729ff5a441b988c69a752a156. The audit performed three model initializations but zero new likelihood evaluations, zero optimizer evaluations, zero sampling evaluations, and zero repetitions of the earlier permutation analysis. All thirteen mandatory final tests passed.

The stable factorial-bilinear representation gives a maximum identity closure error of $2.0940194023211234 \times 10^{-13}$, a maximum spectrum- or multipole-pair aggregation error of -14 , a maximum $|\Delta$

$2m + \Delta$

$2r$) of $4.547473508864641 \times 10$

-13 , and a

$5.479297571220343 \times 10$

maximum parent-functional closure error of $1.6143639203214377 \times 10$

-10 . These satisfy the

original 10

-8 , 10

-8 , 10

-11 , and 2×10

-4 validation thresholds, respectively.

The validation history is retained for reproducibility but not as separate scientific evidence. An initial Q030 execution failed before producing a scientific result because it used a misrecorded common-support hash. The subsequent numerical decomposition in GitHub Actions run 33961469874 produced all nine required diagnostics but failed the -8 for the global identity and 7.19×10

original 10

-8 closure gates at approximately 1.44×10

-8 for grouped aggregation because the calculation formed cancellation-sensitive differences of large quadratic scalars. A later validation-only configuration, run 33961879699, accepted those unchanged values under a 10

-7 numerical threshold but is not

the authoritative final validation. The one-shot precision audit in run 33962896373 then demonstrated that the original 10

-8 gate could be satisfied using algebraically equivalent

stable arithmetic; its overall workflow failure was caused only by a validator schema-key lookup error. The finalization repair did not rerun the scientific audit, change any mask shard, alter any scientific input, or relax any tolerance. The successful run 33963221940 therefore establishes the original-gate result.

These computations all remain internal analyses of the same frozen

Planck/NPIPE/CamSpec likelihood chain. Their progressive agreement strengthens the numerical characterization of that likelihood geometry but does not create multiple independent observational detections. In particular, the exact decomposition does not identify a Planck calibration failure, foreground systematic, defective data vector, unique cosmology-calibration cause, covariance defect, or new physical mechanism
Computational provenance — independent Planck-likelihood portability test.

The subsequent cross-implementation test is recorded as R-CASE031-EDE-PLANCK-PORTABILITY-001. The authoritative execution used q031_planck_portability_v1.py, q031_planck_portability_v1_config.yml, q031_source_lock_v1.json, q031_protocol_lock_v1.json, q031_tests_v1.py, q031_setup_v4.sh, and workflow

.github/workflows/q031-planck-portability-v4.yml in the Morfindien/Bubbleverse repository.

The successful GitHub Actions execution was run 33975028436 on the main branch at head commit ff621d6e0a425a976bb7a55f9194a0de8a87f18f. The environment used Python 3.11.16. The frozen early-dark-energy backend remained mwt5345/class_edc at commit

5a131c91d657dd9a7c6364cc45b038710f8d0d97. The independent Planck likelihood was HiLLiPoP v4.3, component planck_2020_hillipop.TTTEEE, frozen at commit a09ddde3e7ce11df99f74685feb1f1764cafb251.

All nine primary mapped-start optimizations completed. The predefined assessment required tighter continuation, and all nine refinement jobs also completed. The independent-implementation gate and primary completeness gate passed.

The portability decision used the preregistered common-geometry graph classifier inherited by the later Q032 and Q034 analyses. Its clustering coordinates were omega_b, omega_cdm, f_EDE, log10 z_c, theta_i,scf, H_0, and A_planck; endpoints were connected when their normalized common-parameter RMS separation was at most 0.10 and their objective difference was at most 0.50, with repeated support required for a stable cluster.

Under this criterion the resulting stable-basin count was zero and

MATERIAL_STABLE_MULTIBASIN_GATE failed. The downstream direction, primordial, compensation, and residual-support portability extensions were therefore not required for a positive decision and were skipped rather than interpreted as separate failures. The Q031 decision rule consequently returned PORTABILITY_FAILS, final classification The raw HiLLiPoP execution remains valid, but the conclusion of implementation-specific non-portability can no longer be supported by the binary stable-basin result alone. And DOWNGRADE, and geometry classification IMPLEMENTATION-SPECIFIC INTERNAL LIKELIHOOD GEOMETRY, with FINAL_RESULT_GATE passing. This remains the correct historical record of the Q031 computation. Its broader interpretation has subsequently been revised. The Q035 classifier crosswalk applied the same later graph classifier to the preserved historical Q022 CamSpec endpoints and also obtained zero repeated stable basins. The Q031 stable-basin gate therefore does not discriminate the historical CamSpec baseline from HiLLiPoP once the classifier is harmonized.

The final workflow artifact is q031-final-v1, artifact ID 9975019737, with SHA-256 digest a42eac83856d70c9645514916a155b9647fa85f815ce3365771f2f72d4bdca5e. The authoritative validated file is q031_final_validated_v1.json, SHA-256 c4f03c1f1f8701cd14a74173ce2d58a46316eaa00acc86d64ebef4b68d9f4638. The protocol lock has SHA-256 673f0625cb535f302d7439a5b7720c7a14fd8bc8e4fada68c3cf09198c6f16ad, the source lock has SHA-256

56fb6a03cb85b6fd922a9db239c8e343146d2af76efdf2ef15a7e817e7989588, and the locked common-scales file has SHA-256

732aeeb127d9083c0924552aa276117310ca32b5bfda854f51bfd3240638b2f6.

Three earlier workflow versions are retained only as technical provenance. The first failed during environment setup, the second during Python/cache configuration, and the third during ingestion of overlapping Q022 stage and refinement artifacts. None produced a CASE-031 scientific result. The V4 execution is the sole authoritative scientific run.

CamSpec and HiLLiPoP use overlapping Planck PR4/NPIPE observational information. The CASE-031 result therefore represents independent likelihood implementation and computational methodology, not an independent observation of a different sky.

Computational provenance — Q032 common-support bridge and exact-start CamSpec revalidation. The first Q032 TT-only bridge attempt failed during configuration because polarization-only HiLLiPoP nuisance parameters leaked into the native TT construction. It produced no scientific result. The corrected bridge is R-Q032-EDE-PLANCK-TT3PAIR-BRIDGE-002, run Q032-PLANCK-TT3PAIR-COMMON-SUPPORT-BRIDGE-V2. GitHub Actions run 33994305721 completed successfully at execution head

4dc873a5e880d40858d831a3b421456728f0c032. Its exact common TT support contains 5953 dimensions: 143×143 over $\ell = 50\text{--}2000$ and 143×217 and 217×217 over $\ell = 500\text{--}2500$. The support SHA-256 is

f6d7b3195789e7149942c63543bdd8e9a44490ea3f3645fb964e936ec69cca2b. Both CamSpec and HiLLiPoP produced `stable_basin_count = 0` on this restricted surface. The stronger intermediate interpretation that the removed CamSpec information was therefore materially responsible for the historical stable geometry is superseded by the later exact-start control.

A subsequent nested-add-back implementation, R-Q032-EDE-CAMSPEC-ADDBACK-003, GitHub Actions run 34009693922, failed and is retained only as technical provenance. It does not contribute a Q032 scientific result.

The authoritative Q032 execution is R-Q032-EDE-CAMSPEC-ADDBACK-004. The program is `q032_exact_start_addback_v4.py` with configuration

`q032_exact_start_addback_v4_config.yml`, protocol lock

`q032_exact_start_protocol_lock_v4.json`, source lock `q032_exact_start_source_lock_v4.json`,

and workflow `.github/workflows/q032-exact-start-addback-v4.yml`. GitHub Actions run

34015845246 completed successfully at execution head

fbde93aee01ec3bed183207518ff315c96a45c3c.

The V4 protocol converted every sampled parameter reference to a literal scalar value before optimization and audited the exact start vector. The baseline TT3/common-support surface completed all nine profiles and retained `stable_basin_count = 0` after refinement.

The preregistered single-sector add-backs SUPPORT, TE, and EE each completed nine profiles and each retained `stable_basin_count = 0`. The pairwise surfaces SUPPORT + TE, SUPPORT + EE, and TE + EE likewise completed with zero stable basins. The final full-native SUPPORT + TE + EE control also completed nine profiles and refinement with `stable_basin_count = 0`. The mandatory validation and FINAL_RESULT_GATE passed.

The authoritative Q032 classification is

`NO_CLEAN_LOCALIZATION_WITHIN_TESTED_CAMSPEC_INFORMATION_SECTORS`. The raw Q032 computation remains valid: every tested single-sector, pairwise, and full-native surface retained a stable-basin count of zero. The interpretation of why this differed from the historical Q022 classification was subsequently revised. Q032 had originally been treated as replacing a historical non-scalar reference-distribution launch with scalar exact references. The later Q033 provenance audit showed that this distinction did not exist in the executed lineage: the relevant Q022 references were already finite scalars. Q032 therefore supersedes the earlier information-sector localization hypothesis but does not establish scalar-versus-distribution launch semantics as the cause of the historical

discrepancy. It does not invalidate the preserved Q022 endpoint vectors or the subsequent fixed-vector Q023–Q030 calculations. The raw CASE-031 HiLLiPoP execution remains valid, but its cross-implementation portability interpretation is reopened by the later classifier-harmonization result: the historical Q022 CamSpec baseline itself returns zero stable basins when evaluated with the same later graph classifier. The existing Q022, Q031, Q032,

Q033, and Q034 numerical results remain preserved; what changes is the comparative

inference drawn from their classifier labels.

Computational provenance — Q033 launch/reference-semantics audit. The authoritative result is R-Q033-EDE-LAUNCH-SEMANTICS-AUDIT-002, logical run Q033-CAMSPEC-

LAUNCH-SEMANTICS-PROTOCOL-AUDIT-V2.

The audit used `q033_launch_semantics_protocol_audit_v2.py`, `q033_launch_semantics_protocol_audit_v2_config.yml`, `q033_launch_semantics_source_lock_v2.json`, `q033_launch_semantics_tests_v2.py`, and workflow `.github/workflows/q033-launch-semantics-protocol-audit-v2.yml`.

The Q033 V2 files were added to the repository in commit `b03a1ac5dbcb0821039406b099e05982010502fc`, and the workflow was moved into `.github/workflows` in commit `614b227c8de5ea2096f75bf8b97155f340cb1a44`.

GitHub Actions run `34055678308` completed successfully with head SHA `614b227c8de5ea2096f75bf8b97155f340cb1a44`.

The source lock preserves the historical Q019, Q021, Q022, and Q032 implementations needed to reconstruct the relevant parameter semantics. The audit found that all Q022-sampled non-primordial references were scalar after endpoint recentering and that all

Q032 V4 sampled references were likewise scalar. It therefore returned Q033_ANSWER =

NO with classification

`LAUNCH_REFERENCE_SEMANTICS_ALREADY_IDENTICAL_NOT_CAUSAL` and `FINAL_RESULT_GATE = PASS`. No new likelihood optimization was performed. The planned eighteen-shard distribution-versus-scalar comparison was cancelled because its historical distribution-valued arm did not correspond to the actual Q022 execution.

The audit also identified the smallest remaining sampled-parameter difference at that stage. Relative to the fifteen sampled non-primordial Q022 coordinates, the Q032 full-native construction additionally sampled `n_s`, `log A`, and `tau_reio`, whereas Q022 fixed those primordial coordinates at each Q021 vertex.

A subsequent matched Q034 isolation tested this difference directly inside the later full-native construction. Freezing the three primordial coordinates did not recover a reproducibly supported stable basin under the later common-geometry classifier: the refined frozen arm retained `stable_basin_count = 0` and resolved into nine singleton clusters.

The fixed-versus-sampled difference is therefore rejected as sufficient under that later classifier. Its global role in the historical Q022-versus-Q032 comparison remains unresolved

because the historical and later classifications were produced by different mathematical decision rules.

Q022, the Q023–Q030 fixed-vector diagnostics, CASE-031, the raw Q032 V4 result, and the raw Q034 result all remain preserved.

Computational provenance — Q034 primordial-freeze isolation. The authoritative result is R-Q034-EDE-PRIMORDIAL-FREEZE-ISOLATION-001, logical run Q034-PRIMORDIAL-FREEZE-

ISOLATION-V1.

The execution used `q034_primordial_freeze_isolation_v1.py`, `q034_primordial_freeze_isolation_v1_config.yml`, `q034_primordial_freeze_source_lock_v1.json`, `q034_primordial_freeze_tests_v1.py`, and workflow `.github/workflows/q034-primordial-freeze-isolation-v1.yml` in the Morfindien/Bubbleverse repository. GitHub Actions run 34059649759 completed successfully at execution-head commit 8b8035795a9e6064583379afb8663cdc28961764. The frozen physics backend remained `mwt5345/class_edc` at commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.

The existing Q032 V4 FULL_NATIVE sampled-primordial control was reused after identity validation rather than rerun. It contained nine complete decisive profiles and retained `stable_basin_count = 0` and `stable_multibasin = false`. The Q034 test arm changed only the intended primordial parameter status, fixing `n_s`, `log A`, and `tau_reio` to the authoritative values associated with the same nine provenance vectors while preserving the full-native CamSpec construction, scalar-reference semantics, optimizer settings, backend, and numerical thresholds.

All nine Stage-1 jobs completed, the preregistered refinement condition activated, and all nine refinement jobs completed. The decisive frozen arm likewise returned `stable_basin_count = 0` and `stable_multibasin = false`. Its nine refined endpoints formed nine singleton clusters rather than one common collapsed solution.

All mandatory identity, completeness, provenance, freeze, merge, classifier-reproduction, refinement-decision, and interpretation-boundary tests passed, and FINAL_RESULT_GATE passed. The final artifact is `q034-final-v1`, artifact ID 9998302123, with SHA-256 digest 0df519645906dfaaf3b270d9791a74c0f33f25e9fdca8ca344ad2add70f8a87. The source lock records the classifier lineage `q032_exact_start_addback_v4.cluster_rows` — `q032_planck_tt3pair_bridge_v2.bridge_cluster` `q031_planck_portability_v1.cluster_rows`. —

The result is therefore a validated internal computational isolation result, not an independent observation and not a physical detection or falsification of early dark energy. The subsequent classifier-harmonization analysis resolved this interpretive boundary. The historical Q022 stable-multibasin classification used a per-mask mixed cosmology–nuisance geometry, whereas Q031, Q032, and Q034 used the later seven-coordinate common-geometry graph classifier. Applying both definitions to the same preserved Q022,

Q032, and Q034 endpoint sets removes the apparent cross-case disagreement: all three sets are stable under the historical classifier and all three are non-stable under the later

classifier. Primordial freezing therefore remains insufficient to create repeated stable clusters within the later classification framework, but it is neither necessary nor sufficient to explain the former historical-versus-later label difference.

Computational provenance — classifier harmonization. The classifier crosswalk is recorded as R-Q035-EDE-CLASSIFIER-HARMONIZATION-001, logical run Q035-CLASSIFIER-HARMONIZATION-V1. The deterministic artifact-only analysis used `q035_classifier_harmonization_v1.py` together with `q035_classifier_harmonization_v1_config.json`, `q035_classifier_harmonization_source_lock_v1.json`, and `q035_classifier_harmonization_tests_v1.py`. The scientific program files were added to Morfindien/Bubbleverse at commit `b1166e80d5d4c602569da7e32ba02c628dec1975`. No new CLASS calculation, likelihood evaluation, optimization, sampling, seed, or endpoint generation was performed.

The analysis replayed the historical Q022 mixed cosmology–nuisance classifier and the later Q031/Q032 common-geometry graph classifier against the preserved decisive endpoint artifacts. The mandatory result package passed the identity, artifact-only, endpoint-completeness, reference-replay, classifier-crosswalk, final-classification, and physical-safety gates. The final classification is

CLASSIFIER_SEMANTICS MATERIALLY EXPLAINS REPORTED DISCREPANCY.

An initial GitHub Actions workflow invocation failed before the scientific crosswalk because the downloaded Q022 final artifact acquired an additional directory layer and the program could not locate the expected file path. This is a workflow-path failure and carries no physical evidentiary weight. The corrected workflow is `.github/workflows/q035-classifier-harmonization-v2.yml` at revision commit `e12bab3da2972ab2b45ec7d27f569faf5febd025` and includes an explicit artifact-layout gate. A successful V2 GitHub Actions run identifier is not documented in the supplied final result package and is therefore not invented here. This limits CI provenance, but it does not alter the deterministic classifier crosswalk contained in the validated result package.

Computational provenance — continuous cross-implementation geometry. The subsequent common-geometry comparison is recorded as R-Q036-EDE-ENDPOINT-GEOMETRY-PORTABILITY-001. The deterministic artifact-only programme used `q036_endpoint_geometry_portability_v1.py` together with `q036_endpoint_geometry_preregister_v1.json`, `q036_endpoint_geometry_source_lock_v5.json`, and `q036_endpoint_geometry_tests_v1.py`. The authoritative workflow is `.github/workflows/q036-endpoint-geometry-portability-v5.yml` in the Morfindien/Bubbleverse repository.

GitHub Actions run `34105942065` completed successfully on the main branch at execution head `095f90565cf694a7c3383d1fbe736707ab187751`. The run reused the authoritative historical CamSpec refinement endpoints from run `33902262660` and the validated HiLLiPoP final and preflight artifacts from run `33975028436`. No new CLASS calculation,

likelihood evaluation, optimization, sampling, seed generation, or endpoint generation was performed.

The comparison used the seven locked common coordinates ω_b , ω_{cdm} , f_{EDE} , $\log_{10} z_c$, θ_{scf} , H_0 , and A_{planck} and inherited the normalized common-geometry scale definition and 0.10 materiality threshold from the earlier portability framework.

Absolute CamSpec and HiLLiPoP objectives were neither subtracted nor combined.

All nine matched cross-implementation endpoint distances exceeded 0.10, with a median normalized RMS displacement of 2.3623445144. All 36 pairwise geometry drifts likewise exceeded 0.10, with a median of 2.2590148997. The normalized centroid displacement was 1.5370691721. Replay of the preserved HiLLiPoP pairwise reference geometry agreed to a maximum absolute difference of $8.881784197 \times 10^{-16}$. The mandatory identity, provenance, common-coordinate, locked-scale, finite-result, permutation-invariance, reference-replay, no-cross-likelihood-objective, and decision-replay gates passed, and FINAL_RESULT_GATE passed.

The final artifact is q036-final-v1, artifact ID 10012356335. The result is a validated internal computational comparison of two likelihood implementations applied to overlapping Planck PR4/NPIPE information. It supports a material continuous implementation-dependent endpoint difference but does not identify its physical cause, establish an instrumental or calibration systematic, determine which implementation is more accurate, or provide independent evidence for or against early dark energy.

Computational provenance — exact-TT common-support geometry survival. The subsequent deterministic artifact-only comparison is recorded as R-Q037-EDE-TT-COMMON-SUPPORT-GEOMETRY-001. The scientific program is

q037_common_support_endpoint_geometry_v1.py with PROGRAM_ID Q037-TTGEOM-V1.

No CLASS calculation, likelihood evaluation, optimizer, sampler, new seed, or new endpoint generation was performed. Instead, the analysis reused the eighteen authoritative CamSpec and HiLLiPoP refinement endpoints from the exact-TT common-support parent calculation R-Q032-EDE-PLANCK-TT3PAIR-BRIDGE-002, GitHub Actions run 33994305721 at execution commit 4dc873a5e880d40858d831a3b421456728f0c032.

The comparison preserved the seven common coordinates ω_b , ω_{cdm} , f_{EDE} , $\log_{10} z_c$, θ_{scf} , H_0 , and A_{planck} ; the Q031 locked-scale SHA-256

732aeeb127d9083c0924552aa276117310ca32b5bfda854f51bfd3240638b2f6; the common-support SHA-256

f6d7b3195789e7149942c63543bdd8e9a44490ea3f3645fb964e936ec69cca2b; and the materiality threshold of 0.10. Absolute objective values from the two likelihoods were neither subtracted nor summed.

All nine matched endpoint distances exceeded 0.10. The median normalized displacement was 2.9317906965863507 and the maximum was 5.229543793539735. The normalized centroid displacement was 0.8311487237190059. All 36 pairwise geometry drifts also exceeded 0.10, with a median of 1.6367910228418985 and a maximum of

4.755572606658088. Independent result replay returned a maximum absolute difference of 0.0, and the mandatory validation package returned FINAL_RESULT_GATE = PASS.

The result therefore classifies the exact common-support restriction as insufficient to remove the continuous implementation-geometry difference. It weakens unequal TT information support as the primary explanation but performs no causal component attribution. It does not establish a physical Planck systematic, foreground or calibration defect, covariance failure, instrumental problem, early-dark-energy preference or falsification, new physics, or any change in the active H 0 benchmarks.

Computational provenance — exact-TT common-support geometry survival. The deterministic artifact-only comparison is recorded as R-Q037-EDE-TT-COMMON-SUPPORT-GEOMETRY-001. The scientific program is

q037_common_support_endpoint_geometry_v1.py with PROGRAM_ID Q037-TTGEOM-V1. It reused the eighteen authoritative CamSpec and HiLLiPoP refinement endpoints from R-

Q032-EDE-PLANCK-TT3PAIR-BRIDGE-002 rather than running CLASS, evaluating a new

likelihood, optimizing, sampling, changing seeds, or generating new endpoints. The seven common coordinates, the locked-scale SHA-256

732aeeb127d9083c0924552aa276117310ca32b5bfda854f51bfd3240638b2f6, the common-support SHA-256

f6d7b3195789e7149942c63543bdd8e9a44490ea3f3645fb964e936ec69cca2b, and the 0.10 materiality threshold were preserved.

All nine matched endpoint distances exceeded 0.10. The median normalized displacement was 2.9317906965863507 and the maximum was 5.229543793539735. The normalized centroid displacement was 0.8311487237190059. All 36 pairwise geometry drifts also exceeded 0.10, with a median of 1.6367910228418985 and a maximum of

4.755572606658088. Independent result replay returned a maximum absolute difference of 0.0, and FINAL_RESULT_GATE passed.

The scientific workflow is .github/workflows/q037-common-support-endpoint-geometry-v1.yml. GitHub Actions run 34115280043 completed successfully at execution head c49d9885a5bf1e290f4b42f15c59d228c7c036cf. The authoritative raw-endpoint artifact is q037-raw-endpoints-v1, artifact ID 10016050377, with digest sha256:cedc39d5fd7baad93bbf2ba521a5778cd3e69bc768f5a6b977e977e7fa0ed050. The earlier statement that no separate successful Q037 GitHub Actions scientific run identifier was documented is therefore superseded.

Computational provenance — coordinate localization. The subsequent deterministic artifact-only localization is recorded as R-Q038-EDE-COORDINATE-LOCALIZATION-001, with scientific program q038_coordinate_localization_v1.py and PROGRAM_ID Q038-COORDLOC-V1. The workflow is .github/workflows/q038-coordinate-localization-v1.yml. GitHub Actions run 34121000822 completed successfully at execution head 02ddda7c13a53cd7ad20880bf527a86e17ac4d79.

The calculation used the validated Q037 endpoint artifact rather than performing any new CLASS calculation, likelihood evaluation, optimization, or sampling. It preserved the same

seven common coordinates and the locked-scale SHA-256

732aeeb127d9083c0924552aa276117310ca32b5bfda854f51bfd3240638b2f6. All 127 non-empty coordinate subsets were evaluated under preregistered capture and leave-one-label-out robustness rules.

The final classification is DISTRIBUTED. The unique smallest qualifying subset contains ω_b , ω_{cdm} , $f\sigma_8$, and θ_{sc} , with full-sample capture fractions of 90.89% for matched location,

88.48% for centroid displacement, and 87.20% for pairwise shape. All nine leave-one-label-out tests passed in every channel, FINAL_RESULT_GATE passed, and independent result replay agreed to a maximum absolute difference of approximately 1.82×10^{-14} .

The central Bubbleverse program registry records Q038-COORDLOC-V1 as an executable target associated with R-Q038-EDE-COORDINATE-LOCALIZATION-001. This ACTIVE registry status describes the availability of the reproducible program; it does not mean that the scientific question remains open.

The calculation is an internal deterministic reanalysis of the Q032/Q037 endpoint lineage and must not be counted as an independent observational confirmation of the underlying CamSpec-HiLLiPoP difference.

Computational provenance — implementation-block interventions. The subsequent causal-isolation campaign is recorded as R-Q039-EDE-IMPLEMENTATION-BLOCK-INTERVENTION-001. The authoritative execution used PROGRAM_ID Q039-IMPLBLOCK-V5 and workflow .github/workflows/q039-implementation-block-intervention-v5.yml in the Morfindien/Bubbleverse repository. GitHub Actions run 34161368438 completed successfully on the main branch at execution head

365217d4a4ca21e0af3b296c0e6b23df7e07ba60. The scientific configuration preserved the

Q032 exact TT common-support parent, the seven Q031/Q037 common coordinates, the

locked normalization scales, the 0.10 materiality threshold, and the frozen

mwt5345/class_edex backend at commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.

The validated intervention arms tested relative-calibration neutralization, removal of native off-diagonal precision coupling, and their predefined combination. Relative-calibration neutralization produced post-intervention matched-median, centroid, and pairwise-median-drift values of 2.8486368555, 0.8739233292, and 1.4858348360. Removing off-diagonal precision coupling produced 2.6070135030, 0.8411900006, and approximately 1.62217. The combined calibration-plus-precision arm produced 3.0786063935, 1.1944461930, and 1.8740322581. None met the predefined requirement of 0.10 in all three channels, and all three returned zero sufficient leave-one-label-out cases out of nine.

FINAL_RESULT_GATE passed. The result rejects these interventions as sufficient under the tested conditions; it does not establish a physical calibration or covariance defect.

Earlier Q039 V1–V4 executions are retained only as technical recovery provenance. Their failures involved execution, environment, schema, installation, or workflow-placement problems rather than completed scientific tests. They are superseded for scientific interpretation by the successful V5 result and must not be used as physical evidence.

Computational provenance — native foreground profiling freedom. The next matched intervention is recorded as R-Q039-EDE-NATIVE-FOREGROUND-PROFILE-001, with PROGRAM_ID Q039-FGPROFILE-V1 and RUN_ID Q039-NATIVE-FOREGROUND-PROFILE-V1.

The authoritative workflow is [github/workflows/q039-native-foreground-profile-v1.yml](https://github.com/workflows/q039-native-foreground-profile-v1.yml). GitHub Actions run 34184346582 completed successfully on the main branch at execution head 449fe096e49a793446c0c859cf8caf0fcdaf4c8f. All eighteen required optimizer profiles completed, the mandatory result tests passed, the result was sealed, and FINAL_RESULT_GATE passed. The final artifact is q039-fgprofile-final-v1 with SHA-256 digest 5af2927dd93031aeca432866459f380aa368523eb260fc80f9e5a1b79169b93c.

Reference selection was performed separately inside each likelihood; absolute CamSpec and HiLLiPoP objectives were neither subtracted nor summed. The CamSpec foreground reference was taken from label M6-S0, with native sampled coordinates $\text{amp}_{143} = 19.0609711101$, $\text{amp}_{217} = 12.9440793001$, $\text{amp}_{143 \times 217} = 9.8599956054$, $n_{143} = 0.9532159371$, $n_{217} = 1.3084047025$, and $n_{143 \times 217} = 1.3698548767$. The HiLLiPoP reference was taken from M6-S2, with $\text{Aradio} = 60.6375899560$, $\text{Adusty} = 5.9809901364$, $\text{AdustT} = 1.0417701091$, $\text{beta}_{\text{dustT}} = 1.5087185314$, $\text{Acib} = 1.8280064789$, $\text{beta}_{\text{cib}} = 1.7599498553$, $\text{Atsz} = 8.6235873045$, $\text{Aks} = 4.8753015347$, and $\text{xi} = 0.2328374089$. These vectors are implementation-specific native nuisance coordinates and are not interpreted as equivalent physical parameters across the two likelihoods.

After freezing each implementation's own foreground reference vector and reoptimizing the remaining permitted parameters, the full-sample geometry was matched $\text{median} = 2.9829304062$, $\text{centroid} = 1.0554687938$, and $\text{pairwise_median_drift} = 1.6353175394$. The corresponding Q037 baseline values were 2.9317906966, 0.8311487237, and 1.6367910228. All nine leave-one-label-out analyses remained above the predefined materiality threshold, giving a sufficient count of 0/9. The sealed classification is

NATIVE_FOREGROUND_PROFILE-FREEDOM-REJECTED_AS-SUFFICIENT.

The intervention tests native foreground nuisance profiling freedom only. It does not equate the full CamSpec and HiLLiPoP foreground constructions. Full foreground-model harmonization and core likelihood/data-vector harmonization would require coupled changes to multiple methodological layers and therefore cannot be reinterpreted as clean single-block interventions within the frozen Q032 architecture. The resulting unresolved discrepancy remains methodological. Both implementations analyse overlapping Planck PR4/NPIPE information, so these calculations are internal comparative likelihood evidence rather than independent observations of the sky.

External Viability: A Region Is Not a Parameter Vector

The internal likelihood-attribution problem at the active $H_0 = 71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ point is no longer open, and its external viability can now be quantified directly. The external calculation separates three statistical objects that must not be conflated: each dataset's own free- H_0 $n=3$ profile minimum, the same dataset reprofiled with H_0 fixed to $71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and the exact shared cosmological parameter vector of the ACT-supported best-observed basin. Each external chain retains its own nuisance parameters. χ^2 values are not compared across Planck and SPT, and their likelihoods are not summed.

The Planck test uses a public NPIPE CamSpec common-backend approximation motivated by the same- $n=3$ analysis of Efstathiou, Rosenberg, and Poulin; it is not an exact reproduction of their custom paper likelihood. Its free- H_0 profile reaches

-1 Mpc

$H_0 \approx 67.6575 \text{ km s}^{-1} \text{ Mpc}^{-1}$

-1. Reprofiled the same $n=3$ model at $H_0=71.5$ raises the profile objective by

$2 \chi^2(71.5) = +12.873928$.

$\Delta \chi^2_{\text{Planck}}$

The result is stable across the accepted multi-start certification, with a best-two objective spread of approximately 0.40 for the free profile and 0.52 for the fixed- H_0 profile. Planck therefore places substantial pressure on this high- H_0 region within the tested common-backend construction.

-1 Mpc

SPT-3G D1 behaves very differently. Its free- H_0 profile reaches $H_0 \approx 69.624 \text{ km s}^{-1} \text{ Mpc}^{-1}$

-1,

but forcing $H_0=71.5$ while allowing the remaining cosmological and SPT-native nuisance parameters to reprofile costs only

$2 \chi^2(71.5) = +0.342067$.

$\Delta \chi^2_{\text{SPT}}$

The corresponding multi-start spreads are only 0.074 and 0.036 for the free and fixed profiles. A secondary SPT-3G D1+DESI DR2 chain gives an even smaller fixed- H_0 penalty,

$\Delta \chi^2_{\text{SPT} + \text{DESI}}$

$2 \chi^2(71.5) = +0.157387$,

Although this combination cannot be counted as a fully independent confirmation because DESI DR2 is reused elsewhere in the evidence chain. These profile results do not mean that the SPT-only posterior is centred on $H_0=71.5$; rather, they show that the SPT likelihood can accommodate that value with little loss of best-fit quality after the other permitted parameters are adjusted.

The distinction between a high- H_0 region and a particular high- H_0 parameter vector is decisive. When the physically shared parameters of the ACT-supported $H_0=71.5$ solution

are held fixed, while only chain-native nuisance quantities are allowed to adjust, the external penalties become

$\Delta\chi^2_{\text{Planck}}$

$2 = +101.858574$,

$2 = +22.112326$,

$\Delta\chi^2_{\text{SPT}}$

And

$\Delta\chi^2_{\text{SPT+DESI}}$

$2 = +21.511108$.

The exact ACT-optimized vector is therefore not externally portable. SPT can accommodate

$H_0=71.5$, but only by moving to a different region of the $n=3$ EDE parameter space.

Rejecting the exact ACT-supported vector is consequently not equivalent to rejecting every $n=3$ configuration with the same value of H_0 .

This result also clarifies the earlier external literature comparison. Poulin and

-1 Mpc

-1 , but that analysis shares

collaborators find a same- $n=3$ region around $H_0\approx 71\text{ km s}^{-1}\text{ Mpc}^{-1}$

substantial ACT, DESI, and Planck information with the internal chain and is not an

independent replication. Toda and Seto obtain much tighter EDE bounds for an $n=2$

potential, which cannot be transferred directly to the $n=3$ model. Efstathiou, Rosenberg,

and Poulin find strong Planck NPIPE pressure on high- H_0 $n=3$ EDE, while the SPT-3G D1

analysis of Khalife and collaborators does not independently reproduce a strong

preference for $H_0\approx 71.5$. The direct profile calculation now shows why these statements

cannot be compressed into a single verdict on “EDE”: the Planck and SPT likelihoods

respond very differently to the reprofiled high- H_0 region.

The appropriate status of $n=3$ early dark energy is therefore more specific than either

“externally viable” or “ruled out.” The model class remains active, but its ability to sustain

a high inferred expansion rate is strongly dataset- and likelihood-dependent. The Planck

NPIPE common-backend approximation materially disfavors the reprofiled

-1 Mpc

-1 region, whereas SPT-3G D1 can accommodate the same fixed value

$H_0=71.5\text{ km s}^{-1}\text{ Mpc}^{-1}$

with little loss of best-fit quality. Both external CMB families nevertheless strongly disfavor

the exact ACT-supported parameter vector, so the distinction between a high- H_0 region and

one particular high- H_0 parameter vector remains essential.

Figure 5. External tests of the high- H_0 EDE region and the exact shared ACT-supported parameter vector. Each change is measured against the relevant chain's own free- H_0 minimum. Reprofiled and exact-vector evaluation are distinct tests. The chains are not summed; DESI data are reused in the SPT-3G + DESI combination. Values are reproduced from Appendix B.

A covariance-aware attribution of the reprofiled endpoints now resolves the Planck-SPT difference more precisely. The Planck profile penalty of

$$\Delta \chi^2_{\text{Planck}}$$

$$= +12.873928$$

Is not a pure CMB penalty. Within the Planck chain, the combined CMB contribution is only +4.541019: low- ℓ TT contributes -2.274725, low- ℓ EE contributes -0.721566, the NPIPE

high- ℓ TTTEE likelihood contributes +4.273455, and CMB lensing contributes +3.263855. A

further +5.319582 comes from the BAO and Pantheon+ terms, while the chain-native calibration-constraint shapes contribute +3.013327. The total +12.873928 must therefore not be described as though it were a single Planck CMB response.

Within the high- ℓ Planck likelihood itself, the signed full-covariance attribution is strongly non-uniform. TT contributes +4.837082, TE contributes only +0.021571 after substantial internal cancellation, and EE contributes -0.582330. The largest aggregate multipole contribution lies at $500 \leq \ell < 1000$, where the signed attribution is +3.452059. The frequency combinations cancel even more dramatically: the TT 143×217 allocation is +24.634985, while 143×143 and 217×217 contribute -12.927555 and -6.870347, respectively. These quantities are covariance-coupled additive attributions rather than independent χ^2 tests;

2 tests;

individual contributions can therefore be much larger than the final net penalty and can legitimately have either sign.

A later matched-surface calculation makes the localization more controlled. Restricting Planck, SPT-3G, and ACT to the same $n=3$ model, TT+TE+EE observables, and $600 \leq \ell \leq 2000$ interval gives a Planck primary-spectrum increase of

$$\Delta \chi^2_{\text{Planck, primary}}$$

$$= +4.672039$$

$$-1 \text{ Mpc}$$

Between its free endpoint and the reprofiled $H_0=71.5 \text{ km s}^{-1}$

endpoint. Of this

covariance-weighted allocation,

$$+3.224235$$

Lies in $600 \leq \ell < 1000$, compared with $+0.806002$ for $1000 \leq \ell < 1500$ and $+0.641802$ for $1500 \leq \ell < 2000$. The same matched decomposition assigns $+1.906078$ to TT, $+0.861065$ to TE, and $+1.904896$ to EE. These are covariance-coupled allocations rather than independent statistical tests.

The corresponding primary-spectrum changes are dramatically smaller for SPT-3G and ACT: approximately $+0.01465$ and $+0.01512$, respectively. Their internal allocations also differ in sign and structure. SPT gives approximately $+0.02467$ from TT, $+0.26695$ from TE, and -0.27697 from EE, whereas ACT gives approximately -0.01640 , -0.00961 , and $+0.04112$. Thus the small net responses of SPT and ACT are not scaled-down copies of the Planck response.

[Internal computational provenance: R-Q016-EDE-CMB-MECHANISM-004, run Q016-CMB-MECHANISM-V4, GitHub Actions run 33836015653, repository head 91bc0736a00e2454d4e37324e5d3a1a7fe7813d6.]

SPT-3G D1 exhibits a different spectral structure. Its primary TT/TE/EE likelihood contributes $+0.743683$, with TE providing $+1.138190$, TT contributing -0.557279 , and EE contributing $+0.168448$. The multipole dependence again contains large cancellations: the combined contribution is $+0.579083$ for $500 \leq \ell < 1000$, $+0.492192$ for $1000 \leq \ell < 1500$, and -0.863774 for $1500 \leq \ell < 2000$, before returning to positive net contributions at higher

multipoles. The small full-chain SPT penalty,

$$2(71.5) = +0.342067,$$

$\Delta \chi^2_{\text{SPT}}$

Therefore does not mean that its primary spectra are completely insensitive to the high- ℓ region. Rather, the $+0.743683$ primary cost is partly offset by SPT lensing at -0.351681 , by the τ -information at -0.044647 , and by a small Tcal constraint contribution of -0.005287 .

The calibration directions also differ sharply. Planck's $A_{\text{planc k}}$ moves from 0.999733 to 1.004528 , corresponding to approximately 1.92 times its Gaussian width and a constraint-

shape change of $+3.269696$. By contrast, SPT's Tcal moves only from 0.999429 to 0.999493 , approximately 0.018σ . This identifies a substantial Planck calibration direction and a negligible analogous SPT movement, but it does not establish that calibration systematics cause the difference; the nuisance shifts are diagnostic responses within the fitted likelihoods, not causal tests.

The Planck-SPT discrepancy is consequently not a single universal EDE penalty expressed with different numerical strength. This conclusion survives a direct comparison in which Planck, SPT-3G, and ACT are evaluated over deliberately matched observables and multipoles. Across the nine predefined cells formed by TT, TE, and EE combined with the intervals $600 \leq \ell < 999$, $1000 \leq \ell < 1499$, and $1500 \leq \ell < 2000$, none shows the same non-zero $\leq \leq \leq \leq$ residual-shift sign in all three experiments. The tested data therefore provide no support for one common matched residual direction underlying the three likelihood responses. The matched comparison localizes the present discrepancy to experiment-, likelihood-, and covariance-response structure, but localization is not causal identification. Shared cosmic variance has not been incorporated through a unified cross-experiment covariance, no

valid combined Planck–SPT–ACT significance has been calculated, and the analysis does not demonstrate statistical incompatibility between the experiments. Nor does the movement of fitted nuisance parameters by itself establish an unmodelled systematic. The strongest clue remains within Planck: most of its matched primary high- H_0 penalty lies at $600 \leq \ell < 1000$, overlapping the multipole region in which earlier Planck frequency-dependent structure had already appeared.

Planck Likelihood-Architecture Dependence

A subsequent full-multifrequency Planck NPIPE/CamSpec calculation tests that clue directly while preserving the certified cosmological endpoints. The cosmological H_0 is held fixed at the Planck free endpoint, $H_0 = 67.988328967159 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

parameters are held fixed at the Planck free endpoint, $H_0 = 67.988328967159 \text{ km s}^{-1} \text{ Mpc}^{-1}$

and the fixed high- H_0 endpoint, $H_0 = 71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$

and the fixed high- H_0 endpoint, $H_0 = 71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$

while nine Planck calibration and

foreground nuisance parameters are reprofiled with multistart optimization. On this full-multifrequency surface, the fixed-minus-free objective increases by +17.3979348723, while the corresponding raw high- ℓ likelihood contribution changes by +17.2433166263. The larger number must not be treated as a replacement for, or an additive correction to, the matched-surface increase of +5.8238567748: the two calculations use different Planck likelihood constructions. What the new calculation establishes is narrower but important—the Planck high- H_0 penalty remains substantial after the tested calibration and foreground freedom has been reprofiled.

Within $600 \leq \ell < 1000$, the signed covariance-aware TT frequency allocation remains strongly non-uniform. The 143×217 spectrum contributes +18.5085901 and carries 54.79 of the total absolute frequency allocation, while 143×143 contributes -12.9913804, corresponding to 38.46, and 217×217 contributes -2.2803947, corresponding to 6.75. These are covariance-coupled allocations rather than independent χ^2

measurements; the large

positive and negative values therefore represent correlated attribution and cancellation rather than separate statistical detections.

The nuisance-lock tests tell a similarly mixed story. Holding the calibration group to its free-endpoint optimum while reprofiling the remaining nuisance parameters produces the largest atomic lock cost, 9.2694858, equivalent to 45.47 of the normalized non-overlapping lock ranking. The corresponding costs are 3.8495458 for the 143×217 foreground group, 3.6693769 for the 217-GHz foreground group, and 3.5978574 for the 143-GHz foreground group. Calibration is therefore the strongest tested nuisance direction, but the result does not establish that calibration error causes the high- H_0 penalty. Lock costs are diagnostic restrictions on a correlated fit and are not additive χ^2

components. The covariance

decomposition is still less concentrated: the strongest individual target-band covariance pair carries only 17.53 of the absolute pair attribution.

The full-multifrequency result consequently supports neither of two simple descriptions. No individual frequency, nuisance group, or covariance pair is sufficiently dominant to identify a specific Planck direction under the predefined concentration criteria. At the same time, the frequency structure is not sufficiently diffuse to satisfy the predefined

broadly distributed criterion: the largest frequency share exceeds one half, and the effective number of frequency directions is 2.209, below the required value of 2.5. The correct result of this test is therefore indeterminate between specific localization and broad distribution. The evidence is most accurately described as a multidirectional Planck-specific likelihood and covariance response in which 143×217, calibration, the auto-frequency spectra, foreground freedom, and their covariance structure all matter, but no single tested component has been identified as the physical cause.

The scientifically defensible position is therefore narrower than either acceptance or rejection of high- H_0 $n=3$ early dark energy. The model remains active, but its viability is demonstrably experiment- and likelihood-dependent in the CMB analyses tested here. On the deliberately matched surface, Planck supplies the principal resistance to fixing

-1 Mpc

$H_0=71.5 \text{ km s}^{-1}$

-1 , whereas SPT-3G and ACT are nearly indifferent to the same

requirement. The more detailed full-multifrequency Planck calculation shows that the Planck penalty itself also depends substantially on how the Planck likelihood is constructed.

A controlled bridge between the matched CamSpec-lite surface and the full-multifrequency CamSpec surface makes that dependence explicit. At the same certified cosmological endpoints, the matched Planck calculation gives a fixed-minus-free objective increase of $2 = +5.8238567748$,

$\Delta \chi^2_{\text{matched}}$

Whereas the native full-multifrequency construction gives

$2 = +17.3979348723$.

$\Delta \chi^2_{\text{full-MF}}$

These values are evaluations of different likelihood objectives. Their difference,

$17.3979348723 - 5.8238567748 = 11.5740780975$,

is therefore a diagnostic of likelihood-architecture dependence, not an additional physical 2 contribution.

χ^2

The bridge reproduces the native full-multifrequency penalty at 17.3953724212. Removing the native prior-density contribution while retaining the same full-multifrequency likelihood and nuisance space changes the penalty only to 17.1601717140. Objective and prior semantics therefore account for only a small part of the discrepancy between the two Planck constructions. Likewise, fixing the three nuisance parameters shared most directly with the matched treatment— $A_{\text{planc k}}$, $c_{\text{alt E}}$, and $c_{\text{al E E}}$ —while retaining the six multifrequency foreground parameters still leaves a penalty of 15.6176804336. The large full-multifrequency response consequently does not disappear when those shared calibration-like directions are prevented from moving.

The converse reduced-nuisance calculation gives a nominal penalty of 11.7641347631, but that surface is not numerically stable enough for precise attribution: the multistart spread is approximately 5.28 in the fixed- H_0 profile and 6.03 in the free profile. It should therefore be treated as a diagnostic of complicated basin structure rather than as a measured component of the likelihood difference.

Taken together, these tests reject a simple one-component explanation. The additional Planck resistance on the full-multifrequency surface is most accurately described as a coupled likelihood-architecture effect involving the multifrequency data representation, frequency-channel structure, covariance, foreground parameterization, and nuisance coupling. The calculation does not establish that any one of these ingredients is physically responsible, and the corresponding lock and attribution costs must not be added as though they were independent χ

2 measurements. Nor does the result identify a Planck calibration

error, foreground systematic, faulty frequency channel, or new physical effect.

The remaining issue can now be tested without holding the cosmological parameters at the previously certified endpoints. When the same ($n=3$) early-dark-energy model is independently reprofiled on the matched CamSpec-lite and full-multifrequency CamSpec likelihoods, the free minima remain remarkably close in two of the most conspicuous coordinates. The matched likelihood gives

-1 M pc

$H_0=67.9727 \text{ k m s}^{-1}$

$-1, f E D E=0.04627$,

While the stabilized full-multifrequency solution gives

-1 M pc

$-1, f E D E=0.04394$.

$H_0=68.1061 \text{ k m s}^{-1}$

-1 M pc

The differences are only about $(0.13 \text{ k m s}^{-1})$

-1 in (H_0) and (0.0023) in $(f E D E)$. Judged

from these coordinates alone, the two likelihood constructions would appear to prefer almost the same cosmology.

That conclusion does not survive a multidimensional test. Freezing the full-multifrequency preferred cosmology and evaluating it on the matched likelihood after reprofiling the permitted nuisance parameters produces an objective cost of only $(+0.8120)$ relative to the matched likelihood's own optimum. Reversing the test is strongly asymmetric: the matched-likelihood preferred cosmology incurs a cost of $(+13.5564)$ on the full-multifrequency surface. Under the pre-specified operational materiality criterion used for this comparison, a cross-profile cost greater than (1.0) identifies a material shift in the preferred model region. That threshold is a diagnostic convention for likelihood geometry, not a statistical-significance or discovery threshold.

The important result is therefore not a large displacement of (H_0) or $(f E D E)$ themselves. It is that proximity in a few best-fit coordinates does not imply equivalence of the full preferred parameter regions. The Planck likelihood construction changes the multidimensional geometry of the ($n=3$) early-dark-energy solution even when its projected (H_0) and $(f E D E)$ minima remain close. This extends the earlier likelihood-architecture result beyond the

-1 M pc

scalar penalty imposed at $(H_0=71.5 \text{ k m s}^{-1})$.

-1).

The high- (H_0) profiles show the same architecture dependence. After independent

-1 M pc

reprofiling at $(H_0=71.5 \text{ k m s}^{-1})$,

-1), the matched construction incurs an objective

increase of $(+5.0620)$, whereas the full-multifrequency construction incurs $(+15.3993)$.

These reprofiled penalties are not the same statistical objects as the earlier fixed-endpoint

values of (+5.8239) and (+17.3979), and none of the differences between these objectives should be interpreted as an additional physical (χ^2) contribution.

The full-multifrequency free solution has also survived a targeted globality check. Twelve continuation runs across the two leading basins were completed, with a best objective of (10540.9973) and a best-two spread of only (0.0539), well inside the pre-specified stability limit of (1.0). The same validation reproduced the earlier matched-likelihood reference endpoints and their ($H_0=71.5$) penalty to numerical precision. An isolated optimization failure is therefore not a sufficient explanation for the observed cross-profile asymmetry. The origin of the cross-profile asymmetry can now be localized more sharply in parameter space, although its physical or instrumental cause remains unresolved. A controlled four-block hybrid analysis separates the cosmological coordinates into an early-dark-energy timing block ($f_{EDE}, \log_{10} z_c, \theta_{i,scf}$), a primordial block ($n_s, \log A, \tau_{reio}$), a matter block (ω_b, ω_{cdm}), and H_0 itself. For the full-multifrequency CamSpec likelihood, the primordial block dominates the functional response, carrying 71.58 of the absolute Shapley attribution and a signed objective change of +10.5344. The EDE-timing block contributes 17.27 (+2.5424), the matter block 9.15 (+1.3462), and H_0 only 2.00 (−0.2943). The large full-multifrequency response is therefore not primarily an H_0 direction. It is concentrated in a coupled primordial region of the fitted parameter space.

The matched CamSpec-lite surface does not exhibit the same concentrated geometry. Its corresponding response is much smaller and distributed among the matter, primordial, H_0 , and EDE-timing blocks rather than being dominated by the primordial block. The internally reprofiled endpoint span is approximately 0.8632 on the matched surface and 14.1286 on the full-multifrequency surface. These spans should not replace the historical cross-profile costs of +0.8120 and +13.5564, because the newer calculation reprofiles the nuisance parameters at its native endpoints and therefore uses different reference objectives. The absolute foreign-cosmology objectives reproduce the earlier cross-profile evaluations, while the newly reprofiled native endpoints define the geometry used for the block analysis.

The response is also strongly non-additive. For example, the full-multifrequency matter- H_0 pair interaction is −5.7189, demonstrating that the behavior of the likelihood cannot be reconstructed by assigning independent penalties to individual parameter blocks. The Shapley allocations and pair interactions are functional diagnostics of a correlated likelihood surface, not independent physical χ^2 components, and they must not be summed or interpreted as separate observational detections.

This localization can now be resolved one level further. Within the primordial block ($n_s, \log A, \tau_{reio}$), the full-multifrequency response is not localized to a single parameter or to one robust pairwise interaction. Three complete, coherent optimizer restarts all classify the internal structure as distributed. The scalar spectral index n_s is nevertheless the most persistent individual direction: it is the largest single Möbius term in all three full-multifrequency restarts, but its absolute share ranges only from 0.2885 to 0.4891, remaining below the predefined 0.50 threshold for single-component localization. Its

Shapley allocation spans approximately 7.03 to 8.78 on the frozen objective basis. The dominant primordial response should therefore be interpreted as a coupled three-parameter structure in which ns is especially prominent, not as an ns-only effect. The matched CamSpec-lite likelihood also yields a distributed primordial structure in all three coherent restarts, but its detailed decomposition is less stable. The largest term changes from $\log A \times \tau_{reio}$ to $ns \times \log A$ to ns across the three optimizer basins, with dominant absolute fractions of only about 0.1983–0.3665. Comparisons between the full-multifrequency and matched decompositions give total-variation distances from 0.2407 to 0.4980. Because this range crosses the predefined materiality threshold of 0.25, the stronger claim that the internal primordial decomposition is robustly different between the two likelihood constructions is not established. Its detailed composition remains sensitive to the optimizer basin.

The result strengthens the earlier conclusion that the full-multifrequency response is genuinely multidimensional while making its interpretation more precise. It does not identify a calibration error, foreground systematic, instrumental effect, covariance defect, or new physical mechanism. The Möbius and Shapley quantities remain functional diagnostics of the frozen likelihood geometry rather than independent physical χ contributions. The distributed primordial classification also survives every coherent restart and should not be reduced to an ns-only explanation.

Numerically Resolving the Interaction

The exact decomposition can also be evaluated on the preserved numerical likelihood states rather than remaining only an algebraic identity. Using the same preserved historical endpoint vectors, the exact 9,915-element common support, and the fixed full-multifrequency CamSpec precision construction, the interaction was reconstructed for all nine combinations of the three masks and three endpoint-pair edges. No new optimization, sampling, or likelihood evaluation was required for the final audit. These calculations remain valid fixed-vector diagnostics of the historical endpoint family; the later full-native CamSpec revalidation, provenance audit, primordial-freeze isolation, and classifier harmonization narrow the interpretation of that family without changing the arithmetic reported in this appendix. Scalar-versus-distribution reference representation is excluded as the cause of the former stable-basin label difference, and freezing the primordial block does not create repeated stable clusters under the later common-geometry classifier. The subsequent classifier crosswalk goes further: the historical Q022, sampled full-native, and frozen-primordial CamSpec endpoint sets all receive the same classification when a common decision rule is used—stable under the historical mixed classifier and non-stable under the later graph classifier. The former cross-case stable-versus-non-stable discrepancy is therefore classifier dependent. The fixed-vector interaction results reported here remain properties of the preserved endpoint geometry and do not depend on treating those endpoints as classifier-invariant physical basins.

For each diagnostic, define

T P Δ

T D M=-2d

2m

As the data-model factorial coupling and

T Pm)

2(m

T M M=Δ

As the model-quadratic functional term, so that

$$\mathbf{F}=\mathbf{T D M}+\mathbf{T M M} .$$

The resulting signed values are:

Mask 3, edge E01:

$$\mathbf{T D M}_{\hat{i}}+0.1778884724912915, \mathbf{T M M}_{\hat{i}}-0.3555961717667373, \mathbf{F}_{\hat{i}}-0.17770769927544577 .$$

Mask 3, edge E02:

$$\mathbf{T D M}_{\hat{i}}+0.013886599976107456, \mathbf{T M M}_{\hat{i}}-0.027732627515868673, \mathbf{F}_{\hat{i}}-0.013846027539761217 .$$

Mask 3, edge E12:

$$\mathbf{T D M}_{\hat{i}}-0.24647803112273645, \mathbf{T M M}_{\hat{i}}+0.492344754005246, \mathbf{F}_{\hat{i}}+0.24586672288250958 .$$

Mask 6, edge E01:

$$\mathbf{T D M}_{\hat{i}}-0.06892022249060759, \mathbf{T M M}_{\hat{i}}+0.13776448436568042, \mathbf{F}_{\hat{i}}+0.06884426187507284 .$$

Mask 6, edge E02:

$$\mathbf{T D M}_{\hat{i}}+29.235956948609644, \mathbf{T M M}_{\hat{i}}-58.78222077444829, \mathbf{F}_{\hat{i}}-29.54626382583865 .$$

Mask 6, edge E12:

$$\mathbf{T D M}_{\hat{i}}+24.896553819097523, \mathbf{T M M}_{\hat{i}}-50.06464302151693, \mathbf{F}_{\hat{i}}-25.168089202419406 .$$

Mask 7, edge E01:

$$\mathbf{T D M}_{\hat{i}}+2.1572127856770558, \mathbf{T M M}_{\hat{i}}-4.323600159807921, \mathbf{F}_{\hat{i}}-2.166387374130865 .$$

Mask 7, edge E02:

$T D M_{\ell} = -2.1160819949305734$, $T M M_{\ell} = +4.224657002158406$, $F_{\ell} = +2.108575007227832$.

Mask 7, edge E12:

$T D M_{\ell} = -4.042127090605946$, $T M M_{\ell} = +8.056682456803617$, $F_{\ell} = +4.014555366197673$.

The model-quadratic term has the larger absolute magnitude in all nine diagnostics, but this is a statement about the functional decomposition, not about physical causation. In every case the final interaction is produced by the signed combination of both terms. The two largest total interactions remain in the second mask, where edge E02 gives $F = -29.54626382583865$ and edge E12 gives $F = -25.168089202419406$. This preserves the importance of the previously identified $\omega c d m \times A p l$ and k compensation without turning that parameter pair into a uniquely identified physical cause.

The covariance-coupled spectrum decomposition reveals an unusually regular structure.

The $143 \times 217 \rightarrow 217 \times 217$ spectrum pair has the largest absolute contribution to F in all nine diagnostics. For the second-mask E02 interaction alone, this pair contributes

$T D M = -2258.711867910664$,

$T M M = +4519.490435385312$,

And therefore

$F = +2260.778567474648$.

This value is much larger than the final E02 interaction because other spectrum-pair contributions carry opposite signs and cancel it. These pairwise quantities are consequently components of one correlated quadratic form, not independent likelihoods or separately measured effects.

A similarly persistent pattern appears when the same accounting is grouped by multipole.

The $\ell = 1500 - 2000$ self-pair has the largest absolute F contribution in all nine diagnostics.

For the second-mask E02 interaction it gives

$T D M = +8.516985149006956$,

$T M M = -17.14377646828873$,

And

$F = -8.626791319281775$.

The dominant pair therefore lies well above the previously conspicuous $600 \leq \ell < 1000$ region. The earlier intermediate- ℓ structure remains part of the likelihood response, but it cannot be restored as a unique localization of the full interaction.

Because these factorial quantities involve differences between large, nearly cancelling quadratic forms, their numerical representation matters. A direct contrast of the large

scalar quadratics produced closure residuals at approximately the 10

-8 to 10

-7 level.

Evaluating the algebraically identical factorial structure in a numerically stable bilinear form reduces the maximum identity closure error to

$2.0940194023211234 \times 10$

-13,

The maximum spectrum- or multipole-pair aggregation error to

$5.479297571220343 \times 10$

-14,

And the maximum violation of

Δ

$2m = -\Delta$

$2r$

To

$4.547473508864641 \times 10$

-13.

The reconstructed interaction also reproduces the parent common-support functional to within

-10.

$1.6143639203214377 \times 10$

All of these values satisfy the original validation thresholds without relaxing the numerical criteria. The earlier closure discrepancy was therefore a cancellation-sensitive evaluation problem, not a failure of the underlying mathematical identity.

Appendix C — Numerical Search, Endpoint Geometry, Classification and Validation

Historical Endpoint Geometry and Execution Dependence

The full-multifrequency CamSpec calculations produced several strongly separated endpoint families involving both cosmological and nuisance coordinates. Across the preserved historical endpoints, the mean normalized within-vertex RMS separation is approximately 1.61 for the profiled cosmological parameters, 1.90 for the nuisance parameters as a whole, and 1.83 for the multifrequency foreground subset. The cosmology-to-nuisance separation ratio is approximately 0.85, with a bootstrap 16th–84th percentile range of about 0.73–0.99. These quantities remain valid descriptions of the stored endpoint geometry. They do not, by themselves, establish that the same families form stable optimizer basins under every admissible launch procedure.

An important provenance correction concerns how the historical continuation was initialized and classified. The preserved endpoint values were not centres of hidden non-scalar Cobaya reference distributions. Reconstruction of the executed constructor chain shows that the sampled non-primordial parameters already carried finite scalar reference values, recentered directly on the selected endpoint coordinates, while the primordial block `n_s`, `log A`, and `tau_reio` was held fixed at each corresponding primordial vertex. Under the historical continuation, the three restart endpoints were assessed separately within each mask using a mixed geometry containing six profiled cosmological coordinates and nine shared or foreground nuisance coordinates. Cosmological and nuisance RMS separations were calculated separately for each restart pair and combined as

$$D_{\text{joint}} = \sqrt{[(dc_{\text{cosmo}})^2 + (dc_{\text{nuisance}})^2]/2}.$$

A mask was considered collapsed only when both the objective spread was at most 0.50 and the maximum normalized joint separation was at most 0.10. If the three restart solutions remained non-collapsed after the predefined tighter refinement, that mask was classified as stable multibasin structure.

Later full-native CamSpec revalidation also used scalar parameter references

The parameter-direction calculations performed on those preserved endpoint families remain valid as fixed-endpoint geometry. Large normalized components recur among cosmological parameters including H_0 , ω_b , ω_{cdm} , $\log_{10} z_c$, and θ_{scf} ; shared nuisance parameters including A_{planck} , calTE , and calEE ; and multifrequency foreground parameters including amp_{143} , amp_{217} , $\text{amp}_{143 \times 217}$, and n_{143} . Their relative importance changes substantially between the tested vertices. For one representative separation at the first vertex, approximately 89.72% of the squared normalized direction lies in the foreground sector and 9.84% in the cosmological sector. At the second vertex, another separation assigns approximately 73.71% to cosmology, 21.40% to shared nuisance

parameters, and 4.89% to foregrounds. At the third, a representative separation is again mixed, with approximately 47.18% in foreground parameters, 38.98% in cosmology, and 13.84% in shared nuisance parameters. These are geometric diagnostics of the preserved vectors, not probabilities, posterior weights, or causal decompositions.

Nor do the historical endpoint families define a reproducible universal direction across the tested masks. For the three endpoint-pair lineages, the median absolute cosine similarities are approximately 0.190, 0.316, and 0.220. The strongest individual alignment, $|\cos \theta| \approx 0.719$, is not reproduced at the remaining vertex. An exhaustive permutation- and sign-invariant comparison of all 36 joint relabellings likewise recovers no shared edge satisfying the predefined criterion. The best assignment still gives zero shared edges out of three, a global median absolute cosine similarity of approximately 0.3745, and a minimum edge median of approximately 0.3162.

The surviving conclusion is therefore narrower than the earlier basin language implied. The historical full-multifrequency execution generated a multidimensional network of separated cosmology–nuisance endpoint families with mask-dependent local directions, and the subsequent fixed-vector analyses remain valid diagnostics of those stored endpoints.

Later sampled-primordial and matched frozen-primordial full-native executions both returned zero reproducibly supported stable basins under the seven-coordinate common-geometry graph classifier. A direct classifier crosswalk has now applied both operational definitions to the same preserved CamSpec endpoint sets. Under the historical per-mask mixed cosmology–nuisance rule, the historical continuation endpoints, the later sampled full-native endpoints, and the frozen-primordial endpoints are all classified as stable multibasin. Under the later common-geometry graph rule, all three sets are classified as non-stable, with no repeated stable basin. The former stable-versus-non-stable progression therefore disappears when the classifier is held fixed. The historical stable-multibasin label should consequently be understood as a classifier-dependent description of the stored endpoint geometry rather than as a demonstrated classifier-invariant property of the likelihood. This correction also changes the interpretation of the HiLLiPoP portability test: HiLLiPoP still returns no stable basin under the later graph classifier, but the historical CamSpec baseline does the same when evaluated by that rule. The stable-basin gate alone therefore no longer establishes cross-implementation non-portability; whether the underlying CamSpec and HiLLiPoP endpoint geometries differ materially under a harmonized quantitative comparison remains unresolved.

None of these results identifies a foreground failure, calibration error, instrumental systematic, covariance defect, physically distinct cosmologies, or new physics. [Internal computational provenance: historical endpoint continuation R-Q022-EDE-FULLMF-OPTIMIZER-BASIN-002, execution commit 16a7902c3be7d1967adaae9ee380baf0d17e903a; full-native CamSpec revalidation R-Q032-EDE-CAMSPEC-ADDBACK-004, GitHub Actions run 34015845246, execution head fbde93aee01ec3bed183207518ff315c96a45c3c; launch/reference provenance audit R-Q033-EDE-LAUNCH-SEMANTICS-AUDIT-002, run

Q033-CAMSPEC-LAUNCH-SEMANTICS-PROTOCOL-AUDIT-V2, GitHub Actions run

34055678308, head SHA 614b227c8de5ea2096f75bf8b97155f340cb1a44; in the supplied final artifact; frozen backend mwt5345/class_edc commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.]

[Internal computational provenance: historical endpoint continuation R-Q022-EDE-FULLMF-OPTIMIZER-BASIN-002, GitHub Actions run 33902262660, execution commit 16a7902c3be7d1967adaae9ee380baf0d17e903a, program q022_globality_continuation_v2.py; independent HiLLiPoP execution R-CASE031-EDE-PLANCK-PORTABILITY-001, GitHub Actions run 33975028436, execution head ff621d6e0a425a976bb7a55f9194a0de8a87f18f; full-native CamSpec revalidation R-Q032-EDE-CAMSPEC-ADDBACK-004, GitHub Actions run 34015845246, execution head fbde93aee01ec3bed183207518ff315c96a45c3c; launch/reference provenance audit R-Q033-EDE-LAUNCH-SEMANTICS-AUDIT-002, GitHub Actions run 34055678308, head SHA 614b227c8de5ea2096f75bf8b97155f340cb1a44; frozen-primordial isolation R-Q034-EDE-PRIMORDIAL-FREEZE-ISOLATION-001, GitHub Actions run 34059649759, execution head 8b8035795a9e6064583379afb8663cdc28961764; classifier harmonization R-Q035-EDE-CLASSIFIER-HARMONIZATION-001, logical run Q035-CLASSIFIER-HARMONIZATION-V1, program q035_classifier_harmonization_v1.py, program commit b1166e80d5d4c602569da7e32ba02c628dec1975. The corrected V2 workflow is .github/workflows/q035-classifier-harmonization-v2.yml at revision commit e12bab3da2972ab2b45ec7d27f569faf5febd025; a successful V2 GitHub Actions run ID is not documented in the supplied result package. The Q035 calculation is deterministic artifact-only post-processing and introduced no new likelihood evaluation, optimizer, sampler, or scientific dataset. Frozen physics backend: mwt5345/class_edc commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.]

Component and Coordinate Localization

The independently implemented HiLLiPoP calculation remains a valid numerical result. All nine mapped-start optimizations and the required tighter continuations completed, and no reproducibly supported stable basin was found under the preregistered seven-coordinate common-geometry graph classifier. Several HiLLiPoP endpoints nevertheless remained widely separated in common-parameter space despite having similar objective values, so the result should not be confused with convergence to one identical solution.

The original portability interpretation relied heavily on comparing this result with the historical CamSpec endpoint family, which had been classified as stable multibasin structure. That comparison was subsequently shown not to use a common operational definition. The historical CamSpec classification used a per-mask geometry combining cosmological and nuisance-parameter separations, whereas the later analysis used a seven-coordinate common-geometry graph classifier. Applying both definitions to the same preserved CamSpec endpoint sets removes the apparent binary contradiction: all of the tested CamSpec endpoint sets are classified as stable under the historical rule and non-stable under the later graph rule. The stable-versus-non-stable distinction is therefore classifier dependent and cannot by itself establish cross-implementation non-portability.

The surrounding isolation tests remain informative. Restoring the omitted TT support, TE, and EE individually, pairwise, and together did not produce repeated stable clusters under the later classifier. Reconstruction of the executed parameter references showed that both relevant CamSpec paths already used scalar pointlike references, excluding scalar-versus-distribution launch semantics as the explanation. Fixing n_s , $\log A$, and τ_{reio} inside the later full-native construction likewise failed to create repeated stable clusters under that classifier. These results constrain the tested implementations, but they do not restore the historical binary basin label as a classifier-invariant physical property.

The remaining portability question can be addressed without relying on the binary basin label. A deterministic comparison was therefore made directly in the seven coordinates shared by the CamSpec and HiLLiPoP constructions: ω_b , ω_{cdm} , $fEDE$, $\log_{10} z_c$, θ_{scf} , H_0 , and A_{Planck} . Each coordinate was normalized by a scale fixed before the comparison, and the inherited materiality threshold was held at 0.10.

The first continuous comparison revealed a large cross-implementation difference. All nine matched CamSpec–HiLLiPoP endpoint pairs exceeded the materiality threshold, with a median normalized displacement of 2.3623. The normalized centroid displacement was 1.5371. The internal shapes of the endpoint families also differed: all 36 pairwise-distance drifts exceeded 0.10, with a median drift of 2.2590. The historical stable-versus-non-stable distinction was therefore not a valid common-classifier argument, but removing that binary label did not make the underlying fitted geometries equivalent.

A stronger control then asked whether the continuous difference was primarily caused by the two implementations using different temperature-spectrum information. Instead of generating another set of likelihood solutions, the analysis reused the already validated paired CamSpec and HiLLiPoP refinement endpoints obtained on exactly matched TT support. The common support retained the 143×143 spectrum over $50 \leq \ell \leq 2000$ and the 143×217 and 217×217 spectra over $500 \leq \ell \leq 2500$. The same seven coordinates, the same

locked normalization scales, and the same 0.10 materiality threshold were preserved.

The implementation difference survived this restriction. All nine matched endpoint displacements again exceeded 0.10, with a median normalized displacement of 2.9317906966 and a maximum of 5.2295437935. The normalized centroid displacement was 0.8311487237. All 36 pairwise geometry drifts also remained above threshold, with a median of 1.6367910228 and a maximum of 4.7555726067. Independent numerical replay reproduced the derived result with a maximum reported difference of zero.

This materially weakens a simple explanation in which the previously observed displacement is caused mainly by unequal TT information support. It does not show that information support can never matter, but the exact common-support restriction actually tested here does not remove either the matched-location difference or the change in internal endpoint-family shape.

The scientifically defensible conclusion is therefore specific. Under the tested $n=3$ early-dark-energy construction, CamSpec and HiLLiPoP produce materially different fitted parameter geometries, and that difference persists when their TT information support is

explicitly matched. The surviving discrepancy is therefore more plausibly associated with some remaining aspect of likelihood implementation or multidimensional model construction than with the tested support mismatch alone.

A further preregistered localization test asked whether this surviving difference could nevertheless be reduced to a small subset of the seven common coordinates,

$\omega_b, \omega_{cdm}, f_{EDE}, \log_{10} z_c, \theta_{i,scf}, H_0, A_{Planck}$.

All 127 non-empty coordinate subsets were evaluated symmetrically using the same locked normalization scales. To qualify, a subset had to capture at least 80% of the full matched-location, centroid, and pairwise-shape signal and remain robust under leave-one-label-out tests. No one-, two-, or three-coordinate subset satisfied the complete criterion. The unique smallest qualifying subset contained four coordinates,

$\omega_b, \omega_{cdm}, f_{EDE}, \theta_{i,scf}$.

This four-dimensional subset captured 90.89% of the matched-location signal, 88.48% of the centroid displacement, and 87.20% of the pairwise-shape difference. It also passed all nine leave-one-label-out tests in each of the three geometric channels. The surviving CamSpec–HiLLiPoP difference is therefore distributed across a reproducible multivariate parameter structure rather than concentrated in one or two common coordinates.

The decomposition also shows why a simple single-parameter narrative is inadequate.

Different coordinates dominate different aspects of the geometry. The matched-location and pairwise-shape differences receive their largest individual contributions from f_{EDE} , while the centroid displacement is dominated by ω_{cdm} and $\theta_{i,scf}$. H_0 , A_{Planck} , and $\log_{10} z_c$ are not required in the minimal robust subset. This weakens explanations in which the implementation difference is fundamentally an H_0 -only or A_{Planck} -only effect.

The result is nevertheless a localization in fitted parameter space, not a localization of physical causality. It does not show that $\omega_b, \omega_{cdm}, f_{EDE}$, or $\theta_{i,scf}$ is itself responsible for the discrepancy. Possible implementation-level contributors still include foreground treatment, calibration structure, nuisance-parameter semantics, covariance treatment, likelihood construction, or coupled effects among these ingredients.

This remains a methodological result rather than a new observation of the sky. CamSpec and HiLLiPoP analyse overlapping Planck PR4/NPIPE information, so their disagreement cannot be counted as independent observational evidence. The calculation does not determine which implementation is more accurate, establish a Planck calibration or foreground error, demonstrate a covariance defect or instrumental failure, favour or falsify early dark energy, alter the active H_0 benchmarks, or provide evidence for new physics.

[Internal computational provenance: historical CamSpec continuation R-Q022-EDE-FULLMF-OPTIMIZER-BASIN-002, GitHub Actions run 33902262660, execution commit 16a7902c3be7d1967adaae9ee380baf0d17e903a; HiLLiPoP portability execution R-CASE031-EDE-PLANCK-PORTABILITY-001, GitHub Actions run 33975028436, execution head ff621d6e0a425a976bb7a55f9194a0de8a87f18f, HiLLiPoP commit

a09ddde3e7ce11df99f74685feb1f1764cafb251; exact-TT common-support parent R-Q032-EDE-PLANCK-TT3PAIR-BRIDGE-002, GitHub Actions run 33994305721, execution commit 4dc873a5e880d40858d831a3b421456728f0c032, common-support SHA-256 f6d7b3195789e7149942c63543bdd8e9a44490ea3f3645fb964e936ec69cca2b; classifier harmonization R-Q035-EDE-CLASSIFIER-HARMONIZATION-001; continuous geometry comparison R-Q036-EDE-ENDPOINT-GEOMETRY-PORTABILITY-001, GitHub Actions run 34105942065, execution head 095f90565cf694a7c3383d1f736707ab187751, FINAL_RESULT_GATE PASS; exact-common-support geometry result R-Q037-EDE-TT-COMMON-SUPPORT-GEOMETRY-001, program q037_common_support_endpoint_geometry_v1.py, PROGRAM_ID Q037-TTGEOM-V1, GitHub Actions run 34115280043, execution head c49d9885a5b1e290f4b42f15c59d228c7c036cf, FINAL_RESULT_GATE PASS; coordinate-localization result R-Q038-EDE-COORDINATE-LOCALIZATION-001, program q038_coordinate_localization_v1.py, PROGRAM_ID Q038-COORDLOC-V1, workflow .github/workflows/q038-coordinate-localization-v1.yml, GitHub Actions run 34121000822, execution head 02ddda7c13a53cd7ad20880bf527a86e17ac4d79, FINAL_RESULT_GATE PASS. Frozen physics backend: mwt5345/class_edc commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.] Relevant methodological context includes Rosenberg, Gratton & Efstathiou, “CMB Power Spectra and Cosmological Parameters from Planck PR4 with CamSpec,” MNRAS 517, 4620–4636 (2022), arXiv:2205.10869; Efstathiou & Gratton, “A Detailed Description of the CamSpec Likelihood Pipeline and a Reanalysis of the Planck High Frequency Maps,” arXiv:1910.00483; Tristram et al., “Cosmological parameters derived from the final Planck data release (PR4)” (2024), DOI: 10.1051/0004-6361/202348015; Jense, Viña, Calabrese & Hill, “The Choice of Planck CMB Likelihood in Cosmological Analyses,” Physical Review D 113, 043551 (2026), arXiv:2510.09430; and the Cobaya technical implementation.

Testing Candidate Implementation Blocks

Locating a discrepancy in fitted parameter space does not identify the part of the likelihood construction that causes it. The surviving CamSpec–HiLLiPoP difference was therefore subjected to a second level of testing in which specific implementation choices were modified while the exact common TT support, the seven-coordinate geometry, the locked normalization scales, and the predefined materiality threshold of 0.10 were preserved. The first intervention campaign tested relative-calibration semantics and off-diagonal precision coupling. Neutralizing the relative-calibration freedom left a median matched displacement of 2.8486, a centroid displacement of 0.8739, and a median pairwise-shape drift of 1.4858. Removing the native off-diagonal precision coupling gave corresponding values of 2.6070, 0.8412, and approximately 1.6222. Applying both changes together did not produce a cancellation; instead, the three aggregate measures became 3.0786, 1.1944, and 1.8740. None of these interventions approached the required value of 0.10 in all three channels, and none passed the sufficiency criterion in any of the nine leave-one-label-out tests.

Foreground treatment requires a more careful distinction. CamSpec and HiLLiPoP do not use interchangeable foreground parameterizations, so forcing a one-to-one nuisance mapping would not constitute a native matched test. A narrower intervention can nevertheless test whether the freedom of each likelihood's own foreground nuisance parameters to co-profile with cosmology is sufficient to generate the observed geometry. For each implementation separately, the native TT foreground nuisance vector was therefore frozen to the values at that implementation's best authoritative common-support refinement endpoint, while the remaining permitted parameters were reoptimized across all nine labels. No CamSpec nuisance parameter was equated with a HiLLiPoP nuisance parameter.

The discrepancy survived. The post-intervention median matched displacement was 2.9829304062, the centroid displacement was 1.0554687938, and the median pairwise-shape drift was 1.6353175394. For comparison, the corresponding common-support baseline values were 2.9317906966, 0.8311487237, and 1.6367910228. The matched and centroid differences therefore became larger, while the pairwise-shape measure was essentially unchanged. All nine leave-one-label-out analyses remained above the materiality criterion. Native foreground nuisance profiling freedom is consequently insufficient as a single-block explanation of the CamSpec–HiLLiPoP geometry difference. This negative result must not be generalized into the claim that foreground modelling is irrelevant. The intervention tested profiling freedom inside each likelihood's existing native foreground model; it did not make the two foreground models physically or mathematically identical. CamSpec and HiLLiPoP differ more deeply in how foreground treatment is connected to the supplied spectra, detector-set or cross-frequency construction, weighting and coaddition, covariance, masks, and the likelihood data vector itself. Harmonizing those ingredients would necessarily alter several methodological layers simultaneously. A synthetic exchange of data vectors, covariance matrices, or foreground models would therefore cease to be a clean single-block experiment.

The combined result is narrower but stronger than an attribution to any one nuisance choice. No scientifically matchable single implementation block tested here—relative calibration, native off-diagonal precision coupling, or native foreground nuisance profiling freedom—removes the discrepancy, and the tested calibration-plus-precision combination also fails. The remaining explanatory space is concentrated in deeper, structurally coupled differences between the likelihood constructions. This does not demonstrate that a particular coupled mechanism is responsible, nor does it establish that either likelihood is defective or scientifically superior.

Both likelihoods continue to analyse overlapping Planck PR4/NPIPE information. Their disagreement is therefore methodological evidence about likelihood implementation and fitted parameter geometry, not an independent observation of the sky. It does not establish a physical Planck calibration error, an unmodelled foreground contamination, a covariance defect, an instrumental failure, a preference or falsification of $n=3$ early dark energy, or evidence for new physics.

[Internal computational provenance: implementation-block result R-Q039-EDE-

IMPLEMENTATION-BLOCK-INTERVENTION-001, PROGRAM_ID Q039-IMPLBLOCK-V5,

workflow .github/workflows/q039-implementation-block-intervention-v5.yml, GitHub

Actions run 34161368438, execution head 365217d4a4ca21e0af3b296c0e6b23df7e07ba60,

FINAL_RESULT_GATE PASS; native-foreground-profile result R-Q039-EDE-NATIVE-

FOREGROUND-PROFILE-001, PROGRAM_ID Q039-FGPROFILE-V1, workflow

.github/workflows/q039-native-foreground-profile-v1.yml, GitHub Actions run

34184346582, execution head 449fe096e49a793446c0c859cf8caf0fcdaf4c8f, 18/18 profile

jobs complete, FINAL_RESULT_GATE PASS; exact-common-support parent R-Q032-EDE-

PLANCK-TT3PAIR-BRIDGE-002, GitHub Actions run 33994305721, execution commit

4dc873a5e880d40858d831a3b421456728f0c032; common-support SHA-256

f6d7b3195789e7149942c63543bdd8e9a44490ea3f3645fb964e936ec69cca2b; locked-scale

SHA-256 732aeeb127d9083c0924552aa276117310ca32b5bfda854f51bfd3240638b2f6; frozen

physics backend mwt5345/class_edc commit

5a131c91d657dd9a7c6364cc45b038710f8d0d97; HiLLiPoP commit

a09ddde3e7ce11df99f74685feb1f1764cafb251; Cobaya 3.5.6.]

Relevant methodological context includes Rosenberg, Gratton & Efstathiou, “CMB Power Spectra and Cosmological Parameters from Planck PR4 with CamSpec,” MNRAS 517, 4620–

4636 (2022), arXiv:2205.10869; Efstathiou & Gratton, “A Detailed Description of the

CamSpec Likelihood Pipeline and a Reanalysis of the Planck High Frequency Maps,”

arXiv:1910.00483; Tristram et al., “Cosmological Parameters Derived from the Final Planck

Data Release (PR4)” (2024), DOI: 10.1051/0004-6361/202348015; Jense, Viña, Calabrese & Hill,

“The Choice of Planck CMB Likelihood in Cosmological Analyses,” Physical Review D 113,

043551 (2026), arXiv:2510.09430; the official planck-npipe/hillipop implementation at

commit a09ddde3e7ce11df99f74685feb1f1764cafb251; and the Cobaya 3.5.6 CamSpec

implementation.

A Common CMB-Space Test—and Its Numerical Limit

The remaining CamSpec–HiLLiPoP difference raises a difficult methodological question. If

the two likelihoods differ simultaneously in foreground construction, spectrum weighting,

data-vector definition, covariance treatment, and nuisance structure, then simply

exchanging one implementation block for another risks creating an artificial likelihood

that belongs to neither analysis. A cleaner comparison is possible, however, if the common

object is defined physically rather than by forcing the two pipelines to become identical.

For each implementation, the native nuisance parameters can be marginalized separately

while retaining the same underlying CMB temperature spectrum as the shared latent

object. Writing the physical TT spectrum as

$T T, \dots, C \ell_p$

$S = (C \ell_1$

$T T),$

And retaining A Pl anc k as an explicit parameter, the corresponding likelihood can be defined

as

$$CMB(s, A_{Planc k}) \int Li(d|s, A_{Planc k}, \eta_i), \pi_i(\eta_i) d\eta_i,$$

Li

Where i denotes either CamSpec or HiLLiPoP, η_i contains that implementation's own nuisance parameters, and π_i is its native prior. CamSpec data, covariance, foreground modelling, and nuisance priors remain entirely within the CamSpec integral; the corresponding HiLLiPoP ingredients remain entirely within the HiLLiPoP integral. The two likelihoods are neither multiplied nor subtracted from one another. The only common object is the physical CMB TT spectrum.

This construction is methodologically legitimate. Nuisance-marginalized CMB-space likelihoods have an independent precedent in the CamSpec-NPIPE-lite methodology, while the native CamSpec and HiLLiPoP implementations provide the required foreground, calibration, prior, and likelihood definitions. The construction therefore avoids the scientifically problematic alternative of manufacturing a hybrid data vector or covariance matrix.

The first numerical implementation attempted to compress each marginalized likelihood into a single Gaussian representation. That approximation failed its own validation requirements. In CamSpec, the nuisance solution encountered the native upper boundary $\Delta m_{p143}=50$, and the nuisance Hessian was not positive definite. In HiLLiPoP, independently converged nuisance solutions produced a normalized CMB-space spread of 0.2879, well above the predefined stability limit of 0.10. The resulting single-Gaussian Laplace/Schur representation was therefore rejected for scientific use. This is precisely the kind of situation in which ordinary interior Gaussian approximations can become unreliable near parameter-space boundaries or non-Gaussian structure.

A second approach removed the single-Gaussian assumption altogether. The nuisance integral was represented directly using defensive importance sampling and randomized quasi-Monte Carlo. The proposal distribution retained the full native prior as half of the mixture,

$$Q_i(\eta) = \frac{1}{2} \pi_i(\eta) + \frac{1}{6} [Q_{i1}(\eta) + Q_{i2}(\eta) + Q_{i3}(\eta)],$$

With the remaining components centred on three previously defined nuisance solutions.

Because

$$Q_i(\eta) \geq \frac{1}{2} \pi_i(\eta),$$

The corresponding importance ratio satisfies

$$\pi_i(\eta) / Q_i(\eta) \leq 2.$$

This defensive construction preserves sampling support across the entire native prior, including unrecognized modes, while exact truncation respects native hard boundaries. Hesterberg's defensive-mixture framework provides the statistical motivation for this bounded-weight construction, while Owen-scrambled Sobol sequences provide the randomized quasi-Monte Carlo basis for independent numerical replicates.

The numerical campaign used four independent scrambled replicates for each likelihood and an explicitly finite sequence of sample levels from $m=9$ through $m=16$, reaching a hard cap of 393,216 nuisance nodes per replicate and implementation. Convergence was assessed from changes in relative log-likelihoods across the preserved probe set and from agreement between two independent pairs of replicates. Both tests required differences no larger than 0.05.

The full finite campaign completed, but the required stability was not obtained. At the hard cap, the maximum transition change in the CamSpec probe likelihood was 16.59, and the corresponding difference between the two independent replicate banks was 28.44. For HiLLiPoP, the values were still larger, at 61.13 and 63.11, respectively. All are far above the predefined tolerance of 0.05.

The consequence is methodological rather than physical. Because the nuisance marginalization did not pass numerical validation, the subsequent cosmological endpoint calculations were deliberately not executed. The analysis therefore provides no validated value for the matched displacement, centroid displacement, pairwise-shape drift, or leave-one-out geometry under the marginalized likelihoods. It neither establishes that the common-CMB construction would remove the CamSpec–HiLLiPoP discrepancy nor establishes that the discrepancy would survive a successfully validated implementation. What has been learned is narrower but important. A scientifically meaningful common latent CMB likelihood can be defined without constructing a synthetic hybrid likelihood, but neither the tested single-Gaussian compression nor the finite defensive-RQMC implementation was numerically adequate for the required comparison. The surviving implementation-dependent geometry therefore remains unresolved at the deeper construction level. This is not evidence that either Planck likelihood is defective, does not favour one implementation over the other, and does not constitute evidence for or against new physics.

An important external qualification is required. Jense et al. (2026) found that, for the Planck multipole range retained when Planck is combined with ground-based CMB observations, differences between Planck map and likelihood products become small, with extended-model constraints becoming insensitive to the Planck product choice in the combinations they tested. This does not directly contradict the Bubbleverse CamSpec–HiLLiPoP fitted-geometry result because the tested model, likelihood construction, and combined data are not identical. It does, however, prevent the internal geometry difference from being promoted to a claim of materially different cosmological inference until that consequence is demonstrated in a matched downstream analysis.

Methodological sources: Rosenberg, Gratton & Efstathiou (2022); Efstathiou & Gratton (2021); Tristram et al. (2024); Jense, Viña, Calabrese & Hill (2026); Self & Liang (1987); Spokoiny (2023); Hesterberg (1995); Owen (1998).

[Internal computational provenance: single-Gaussian compression result R-Q040-EDE-CMBSPACE-COUPLED-BRIDGE-005, classification COMPRESSION_VALIDATION_FAIL; mathematical construction R-Q040-MATH-DEFENSIVE-MARGINAL-001; defensive-RQMC

result R-Q040-EDE-DEFENSIVE-RQMC-CMB-MARGINAL-004, PROGRAM_ID Q040-RQMC-V4, GitHub Actions run 34258155387, workflow .github/workflows/q040-defensive-rqmc-v4.yml, execution head 855a28c58246dbdd52d01cae1f40e0611103d96b, final artifact q040-rqmc-final-v4, SHA-256

450c7a4116eff544d5c8e0f8b209a6baf41caa4cb8b89284923d1956b348a62; frozen physics backend mwt5345/class_edc commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97; HiLLiPoP commit a09ddde3e7ce11df99f74685feb1f1764cafb251; Cobaya 3.5.6.]

Downstream Scientific-Consequence Portability and the V19 Convergence Limit

After the common-CMB-space marginalization programme failed numerical validation, Bubbleverse attempted a direct downstream inference test of the surviving CamSpec-HiLLiPoP fitted-geometry difference. The authoritative executed implementation is Q041-PLANCKPORT-V19 with result identifier R-Q041-EDE-DOWNSTREAM-PORTABILITY-019. The scientific question encoded by V19 was whether the validated $n=3$ EDE fitted-geometry difference produces materially different cosmological inference when the CamSpec and HiLLiPoP Planck arms are combined separately with the same V19 external CMB and late-time information. The arms were never multiplied together, and absolute cross-arm likelihood or χ^2 subtraction remained forbidden.

The V19 matrix contains eight data combinations for each Planck arm: P, P_A6, P_L6, P_D2, P_A6_L6_D2, P_L6_D2, P_A6_D2, and P_A6_L6. A6 is ACT DR6 foreground-marginalized primary CMB; L6 is ACT DR6 lensing; D2 is DESI DR2 BAO. Two independent chains were required for every arm/combination pair, giving 32 logical chains.

The frozen source lock retains the Q032 exact-common-TT parent, class_edc commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97, HiLLiPoP commit a09ddde3e7ce11df99f74685feb1f1764cafb251, Cobaya 3.5.6, ACTCollaboration/DR6-ACT-lite v1.0.1 commit 880eacb40d66722eb1c32d7b5621e91662b4d808, ACTCollaboration/act_dr6_lenslike v1.2.1 commit b386ddb5821c1216c709f051c9289292f174d30, and DESI DR2 through bao_data v2.6 commit b7b8a36e9bccb063081f811f323cada21ab5fbdd with Cobaya definition commit b76b6fed2a6c8c5594c6f92d5058bef10079746a.

When A6 is absent, V19 uses the full authoritative Q032 exact-common-TT support. When A6 is present, Planck primary TT is further restricted to $\ell \leq 599$ and ACT primary TT/TE/EE $\leq$ begins at $\ell = 600$. Cross-likelihood covariance is assumed zero only after that disjoint primary-multipole construction. This is the V19 overlap policy; it is not an official PlanckActCut claim.

The sampler is Cobaya MCMC with $R_{\text{minus1_stop}} = 0.02$, $R_{\text{minus1_cl_stop}} = 0.2$, two independent chains per arm/combination, $\text{max_samples} = 20,000$ accepted samples per chain, and a separate hard science R-hat gate of 1.05. Eight successful compute segments per chain are permitted. Each segment has a 300-minute soft stop and a 330-minute GitHub job timeout.

V19 uses a fixed linear eight-stage workflow. Each segment transports the exact immediate-predecessor chain state through same-run GitHub artifacts. Cross-version chain-state reuse is forbidden. HiLLiPoP segmented resume is locked to the exact serialized `planck_2020_hillipop.TT` component from `chain.updated.yaml`; `Cobaya --allow-changes` is not used.

The final execution did not obtain the required posterior convergence. Thirty of the 32 logical chains entered `MAX_SAMPLES_WITHOUT_CONVERGENCE`. The two remaining CamSpec chains, `P_A6/c1` and `P_A6_L6_D2/c1`, reached segment 8 still in `PARTIAL` state after all eight compute segments. The combined record therefore contains zero logical chains with documented `COMPLETE` status.

The authoritative final object records `scientific_classification = NO_SCIENTIFIC_RESULT`, `outcome_type = CONTROLLED_NO_SCIENTIFIC_RESULT`, `no_science_reason = MAX_SAMPLES_WITHOUT_CONVERGENCE`, `actual_computed_result = false`, `technical_failure = false`, `final_outcome_valid = true`, and `status = PASS`.

`CHAIN_COMPLETENESS`, `ALL_RHAT_LE_1P05`, and `BOTH_ARMS_ALL_COMBINATIONS` are

BLOCKED; Q040_FIREWALL = PASS; SEGMENTED_RESUME_COMPLETENESS = CONTROLLED_STOP.

The green terminal state is therefore a validated control outcome, not a converged science result. The comparator never acquires a valid complete matrix from which to compute an authoritative material, equivalent, or constrained-mixed portability class. Non-convergence cannot be used as physical evidence for agreement or disagreement between CamSpec and HiLLiPoP.

A contract audit exposes a second, independent limitation. The broader downstream design preceding V19 required both Λ CDM and $n=3$ EDE in each Planck arm, a common supernova likelihood in addition to ACT primary, ACT lensing, and DESI DR2, four leave-one-out combinations, within-arm Λ CDM-versus-EDE model-preference information, and a different materiality rule. V19 is an EDE-only A6/L6/D2 test with three leave-one-out combinations and a Gaussian Bhattacharyya summary in its own classification logic.

This scope difference does not retroactively invalidate the preregistered V19 experiment. It means only that V19 and the broader original design are not the same scientific test. Even a converged V19 result would not by itself document the omitted Λ CDM control, supernova dependence, fourth leave-one-out arm, or original model-preference criterion.

The downstream scientific consequence of C-036-IMPL therefore remains unresolved. Any new attempt should be registered as a new execution configuration rather than increasing V19 limits after seeing its outcome. Existing V19 chains, progress histories, checkpoints, proposal adaptation, boundary behaviour, and acceptance diagnostics are legitimate inputs to a new numerical-design study, but the V19 scientific identity and terminal result must remain unchanged.

Internal computational provenance: repository Morfindien/Bubbleverse; program Q041-PLANCKPORT-V19; result R-Q041-EDE-DOWNSTREAM-PORTABILITY-019; workflow `.github/workflows/q041-planck-portability-v19.yml`; GitHub Actions run

34979609004, attempt 4; execution head a962ba70077422f8b69642dea4e8272e1d00e5ff;
final artifact q041-final-v19, artifact ID 10754769330, SHA-256
2284518ecbaaf1ff0d0093977827d5162ec29549ba2b81880f266c250311ca2b;
q041_final_v19.json SHA-256
0062874ed3529004f46cfb542a708d5b8e8dde55d1b4fd1c2ac7c394e6c8be79;
q041_final_tests_v19.json SHA-256
95fe1e63c78033de8559f609f0d8a1f4d48ee1b4123f3bb8f845b3ac6e5797ab.

This appendix preserves the detailed numerical material behind the optimizer, endpoint, classifier and cross-implementation analyses summarized in the main book. It contains the historical endpoint geometry, execution-dependent basin classifications, classifier harmonization, coordinate localization, tested implementation-block interventions, the common-CMB-space marginalization programme, and the subsequent downstream portability campaign and convergence limit. These records are preserved at technical depth because their methodological details are necessary for audit and reproduction, but they are not required to follow the main scientific narrative.

Appendix D — Bubbleverse-Model Repository

D.1 Purpose of the Repository

Morfindien/bubbleverse-model is the version-controlled repository for the accepted, candidate and formal Bubbleverse model state. It is distinct from Morfindien/Bubbleverse, which is the principal scientific source repository containing research programmes, numerical experiments, results, manuscript material and Q-history.

The model repository is designed as a controlled compression of the scientific state that Bubbleverse has accepted inside an authorized Q boundary. It is not presented as established physical truth. Its purpose is to make the current state explicit, machine-readable, testable, versioned and traceable.

Repository snapshot inspected for this appendix: 9 September 2026.

D.2 Current Repository State

The inspected repository has main as its active branch. The inspected main-branch head is e1bd4ae27cd3e5a2172bc5e941790c4f2d091643, a repository-cleanup and metadata-synchronization commit.

The accepted scientific model is v0.2, model revision R000002, with authorized scientific boundary Q001–Q040. The accepted state records Q040 as the current processed question.

The accepted state derives from source repository Morfindien/Bubbleverse at commit 855a28c58246dbdd52d01cae1f40e0611103d96b and records the Word input bubblevers 0.40m.docx with SHA-256

d389095fea2bd65ca5424a77cf22a4240581f1966eff3575ddabddfd0dd54397.

The Q040 accepted-state note is deliberately conservative. Q040 provides a methodologically legitimate common physical CMB-space nuisance-marginalization construction for CamSpec and HiLLiPoP, but the tested numerical representations failed mandatory stability validation. No marginalized endpoint geometry was therefore accepted. Technical failure is not treated as physical falsification, and the causal origin of the implementation-geometry discrepancy remains unresolved.

A later source-repository execution, Q041-PLANCKPORT-V19, occurred after the accepted-model snapshot described here. It produced a validated controlled no-science computational outcome rather than an accepted physical model change. Nothing in that run authorizes silently extending the inspected model-repository boundary beyond Q040; its provenance is documented in Appendix C and in the Q-Journal.

The later execution also revealed a scope divergence between the V19 preregistration and the broader original downstream design. That divergence is relevant to future model

promotion because a controlled no-science result from the narrower V19 matrix does not constitute completion of the broader Λ CDM/EDE plus supernova comparison.

D.3 Repository Architecture

The repository root is organized into distinct scientific and operational layers.

Accepted contains the current accepted scientific state.

Candidate contains the proposed or most recently promoted candidate state and its retained promotion record.

Model contains the machine-readable formal model and its schemas.

Evidence contains the evidence registry and Q-specific update material.

Provenance contains source and Q provenance records.

Tests contains test definitions, preregistration material, test programmes, results and the test registry.

Versions contains frozen accepted-model snapshots.

Release contains the canonical internal release handoff.

Changelog contains human-readable and machine-readable model history.

.github contains the public workflow and repository-maintenance automation.

Bubbleverse_model_program_registry.json provides a central programme registry for callable model functions.

D.4 Canonical Accepted Model

The canonical accepted state is not represented by one scientific prose file alone.

Accepted/model_state.json is the central state pointer. It identifies accepted model v0.2, revision R000002, Q001–Q040, and maps the accepted scientific layers to:

Accepted/constraints.json

Accepted/contradictions.json

Accepted/mechanisms.json

Accepted/observations.json

Accepted/predictions.json

Accepted/robustness.json

Accepted/MODEL_CURRENT.md supplies the corresponding human-readable accepted-state summary.

The accepted directory therefore functions as the canonical current scientific layer, while model provides the formal machine-readable model description and versions/accepted preserves frozen historical accepted states.

D.5 Formal Model Layer

The model directory contains the formal and computational description of the model rather than another copy of the reader-facing manuscript.

Its inspected files include assumptions.json, benchmarks.json, domain_of_validity.json, environment.json, equations.json, external_catalog_registry.json, input_schema.json, limitations.json, model_manifest.json, output_schema.json, parameters.json, result_registry.json and uncertainty.json.

Together these files separate model assumptions, equations, parameters, benchmarks, uncertainty, execution environment, interfaces, registered results, domain of validity and explicit limitations.

This separation is important. A value appearing in a benchmark registry is not automatically promoted into an unrestricted scientific truth claim. The repository remains Q-bounded and status-aware.

D.6 Accepted and Candidate States

The repository uses an accepted/candidate workflow.

New completed Q material first enters an authorized evidence and model-diff process. A candidate state is built before accepted state is changed. Formal synchronization, scientific tests, formal tests, regression and integrity tests, and a promotion gate occur before a successful candidate becomes accepted state.

The retained Q040 candidate record identifies candidate v0.2 / R000002, previous accepted state v0.1 / R000001 at Q039, source Bubbleverse commit

855a28c58246dbdd52d01caef40e0611103d96b, and promotion status PROMOTED_TO_v0.2.

Candidate files therefore remain useful provenance after promotion. Their existence must not be interpreted as evidence that a second scientific state is simultaneously active.

D.7 Version History

The repository preserves accepted snapshots under versions/accepted.

At the inspected state, two accepted-version directories exist:

Versions/accepted/v0.1

Versions/accepted/v0.2

The machine-readable change registry records the principal evolution.

The repository began with a Q039 v0.1 candidate bootstrap. That candidate passed its scientific campaign and was promoted to accepted v0.1. Subsequent non-scientific integrity repairs separated accepted and candidate paths, strengthened Git provenance and future promotion logic, and installed the formal-model layer.

Later Q039 structural additions installed the result-context and next-step router and external astronomy-catalog context without changing the accepted scientific conclusions.

Q040 produced accepted v0.2 / R000002. Subsequent entries record a post-promotion

metadata synchronization and repository cleanup explicitly marked as non-scientific changes.

D.8 Evidence and Provenance

The repository distinguishes scientific source history from model state.

Morfindien/Bubbleverse remains the scientific source repository.

The model repository contains evidence/evidence_registry.json and Q-specific update records including Q039 and Q040 material. The provenance directory contains dedicated

Q039 and Q040 provenance documents together with bootstrap provenance.

The intended traceability chain is:

Q to source to evidence to programme or result to candidate change to tests to promotion to accepted version to Git commit.

Where run Ids, artifact hashes, programme Ids, result Ids or source commits are available, they may be attached to that chain.

The repository was created late in the Q001–Q040 research sequence. Consequently, the complete historical execution record for early Qs does not originate natively inside this repository. Earlier scientific provenance remains primarily in the Morfindien/Bubbleverse repository, the manuscript and the Q-Journals.

D.9 Testing and Regression Protection

The tests directory contains TEST_PLAN_CURRENT.md, preregistration material, programmes, result records and test_registry.json.

The model's public documentation distinguishes several levels of validation. The current repository documentation reports a nine-test scientific campaign, twelve formal-model validation checks and a twenty-check public system healthcheck. The Q040 release handoff separately records eleven mandatory model-update tests passed, with no blocked, failed, inconclusive, invalid, not-comparable or technical-failure mandatory tests.

The tests protect scientific invariants including Q-boundary control, H_0 inference-chain identity, contradiction preservation, separation of technical from scientific failures, open-question preservation and JSON/schema integrity.

Regression protection is therefore not merely a software-health mechanism. It is also intended to prevent later model updates from silently deleting unresolved contradictions or converting technical failures into scientific conclusions.

D.10 Current Test-Metadata Inconsistency

One repository inconsistency must remain visible.

Tests/test_registry.json identifies accepted model v0.2, current Q040 and Q001–Q040 access, but several descriptive why_test_matters strings still refer to Q039 as though Q039 were the candidate boundary. The same file nevertheless marks its registered tests PASS.

The canonical release handoff independently identifies Q040, v0.2, R000002, eleven mandatory tests passed and repair status

Q040_POST_PROMOTION_CONSISTENCY_REPAIRED.

The stale Q039 wording should therefore be treated as a repository metadata inconsistency requiring maintenance. It is not evidence that the scientific model reverted to Q039, and it must not be silently rewritten here as though the repository conflict did not exist.

D.11 Workflow Architecture

The permanent public entry workflow is:

.github/workflows/00-bubbleverse-model-start-public.yml

The public interface exposes non-destructive status and validation operations together with bounded computational functions.

Documented operations include test, validate-model, model-status, contradictions, provenance, release-handoff, gaussian-tension, difference, percent-shift, weighted-mean, known-results, contextualize-result, astronomy-catalog-context and campaign.

The public test and status functions are not scientific-promotion commands. The repository documentation explicitly separates ordinary execution and validation from the controlled model-update pipeline.

A repository-cleanup workflow also exists as supporting infrastructure. Cleanup or metadata synchronization must not be confused with scientific model revision.

D.12 Q Firewall

Bubbleverse-model is explicitly Q-bounded.

At the inspected state, the authorized range is Q001–Q040.

Evidence beyond Q040 must not influence scientific claims, parameters, equations, assumptions, contradictions, mechanisms, predictions, calculations, benchmarks, result registries or candidate revisions until the boundary is expanded through an authorized model update.

The later Q041 computational result is therefore treated here as post-boundary provenance, not as accepted-model state. No Q041 promotion record is established in the supplied model-repository snapshot, and no physical model change follows from the controlled no-science outcome.

The firewall is therefore a scientific-state constraint, not merely a file-naming convention.

D.13 Release and Promotion

The repository does not rely on GitHub Releases as the canonical scientific release mechanism at the inspected state. The canonical handoff is stored under `release/MODEL_RELEASE_HANDOFF.json` and its human-readable companion.

The Q040 handoff records promotion from accepted v0.1 to v0.2, candidate promotion PASS, contradiction PASS, current-Q PASS, final-audit PASS, high-Q-contamination PASS, input-state PASS, model-schema PASS, provenance PASS, Q-firewall PASS and regression **PASS**.

The release state is `ALL_GREEN`.

The recorded model-repository commit `970344fc1aa4ff1b857f61847f14119ba3f0ead2` is identified specifically as the promotion-input candidate commit. Later repository consistency repair, metadata synchronization and cleanup commits therefore belong to the repository history after that promotion point.

D.14 Reproducibility Classification

CURRENT MODEL STATE RECONSTRUCTABLE.

The canonical accepted version, revision, Q boundary, accepted-layer paths, source Bubbleverse commit, Word input hash, candidate state, release handoff and accepted snapshots are explicitly recorded.

MODEL VERSION HISTORY PARTIALLY RECONSTRUCTABLE.

The v0.1 to v0.2 transition is documented, together with structural repairs and formal-layer installation. The model repository itself, however, began near Q039 and does not contain native per-Q model-state history for the complete Q001–Q038 period.

TEST ARCHITECTURE RECONSTRUCTABLE WITH METADATA CAVEAT.

The principal test directories, registries and release gates are available, but stale Q039 wording remains in the Q040 test registry.

WORKFLOW ARCHITECTURE RECONSTRUCTABLE.

The permanent public workflow and its documented operation classes are available.

SCIENTIFIC EXECUTION HISTORY PARTIALLY RECONSTRUCTABLE FROM THIS REPOSITORY ALONE.

The complete experiment history belongs primarily to Morfindien/Bubbleverse and the Q-Journal record.

D.15 Relationship Between the Three Records

The Bubbleverse research system now has three distinct archival layers.

Morfindien/Bubbleverse records the scientific work: programmes, workflows, experiments, artifacts and research outputs.

Morfindien/bubbleverse-model records the accepted and candidate scientific model state, formal representation, tests, promotion history and model provenance.

The Bubbleverse Q-Journals record the chronological human-readable decision trail.

These records overlap deliberately but are not interchangeable.

The research repository answers what was run.

The model repository answers what scientific state was accepted and how it is versioned.

The journals answer how the investigation moved from one question to the next.

D.16 Known Repository Gaps

The Q001–Q038 model history predates the dedicated model repository and is therefore not represented by native historical accepted snapshots for every Q.

Several Q040 test-registry descriptions retain stale Q039 wording despite the current Q040 identity.

Complete tag history was not established during this reconstruction.

The absence of a historical record in one repository must not be filled by inference from adjacent files. Where provenance is missing, it remains missing until independently reconstructed.

A post-Q040 scientific-source execution now exists in Morfindien/Bubbleverse, but the supplied Bubbleverse-model snapshot contains no corresponding accepted-model promotion beyond Q040. Future updates must preserve that distinction rather than treating source-repository execution as automatic model acceptance.

The executed V19 downstream test also differs materially in scope from the broader original Q041 design. Any later promotion or successor execution should record which contract is being tested and must not merge the two designs into one retrospective result.

FIG-V1-002

PRIORITY:

MANDATORY

FIGURE LOCATION:

Chapter 2 – immediately after “A Continuum of Inference”.

FIGURE ACTION:

CREATE PROVENANCE / INFERENCE-CHAIN DIAGRAM.

FIGURE PURPOSE:

Show why different H_0 estimates cannot be treated as identical measurements.

FIGURE TYPE:

Scientific network schematic.

FIGURE PROVENANCE:

Chapter 2 surviving H_0 route descriptions. B

Bubblevers 0.30m.docx

FIGURE PRODUCTION PROMPT:

TITLE OF IMAGE: Independent and Partially Shared Routes to H_0

TYPE: Publication-style scientific inference network.

PURPOSE: Show the principal routes that infer the same parameter H_0 while using different observations, calibrations, and model dependencies.

CONTENT: Place H_0 at the right or centre as the shared inferred parameter.

Create separate pathways for:

Local Distance Network Geometric anchors → stellar calibrators → supernovae or Hubble-flow tracers → finite-redshift distance relation → H_0

Megamasers Disk positions + velocities + accelerations → geometric galaxy distances + peculiar-velocity treatment → H_0

Strong-Lensing Time Delays Image configuration + time delays + lens mass model + stellar kinematics + line-of-sight structure → time-delay distance → H_0

Gravitational-Wave Standard Sirens Waveform distance + host/statistical redshift + selection and inclination modelling → H_0

BAO + BBN BAO geometry + sound-horizon calibration + baryon-density information + cosmological framework → H_0

CMB TT/TE/EE spectra + recombination/acoustic physics + cosmological model → H_0

Use dotted or visually distinct shared-dependency connectors only where the manuscript explicitly states overlap, especially among local calibrators.

REQUIRED VALUES: H_0 Distance Network: $73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$ Megamasers: $73.9 \pm 3.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ TDCOSMO cited primary analysis: $71.6^{+3.9}_{-3.3} \text{ km s}^{-1} \text{ Mpc}^{-1}$ DESI DR2 + BBN: $68.51 \pm 0.58 \text{ km s}^{-1} \text{ Mpc}^{-1}$ Combined Planck + SPT-3G + ACT DR6 under flat Λ CDM: $67.24 \pm 0.35 \text{ km s}^{-1} \text{ Mpc}^{-1}$

Do not force broad standard-siren intervals into a compact central-value comparison if doing so hurts readability; they may be labelled “currently broad” in this diagram and plotted precisely in FIG-V1-003.

LAYOUT: Landscape network flowing toward H_0 .

VISUAL STYLE: Technical scientific schematic.

BACKGROUND: Light or transparent.

ASPECT RATIO: 16:9.

SCIENTIFIC RESTRICTIONS: Do not call all pathways independent. Do not imply that H_0 is a direct instrument reading. Do not merge BAO+BBN and CMB into one route. Do not remove model/calibration dependencies.

FINAL GENERATION INSTRUCTION: Create a publication-quality inference-chain map showing the physically distinct and partly overlapping routes by which the manuscript infers H_0 .

FIGURE CAPTION COPY BOX:

Figure 2. Principal inference routes to H_0 . The same cosmological parameter is reached through physically and statistically different chains involving geometric calibration, stellar indicators, gravitational lensing, gravitational waves, standard-ruler physics, and CMB modelling. Their different dependencies are central to the interpretation of agreement and disagreement among H_0 estimates.

FIG-V1-006

PRIORITY:

HIGH VALUE

FIGURE LOCATION:

Chapter 4 – “Planck Likelihood-Architecture Dependence”.

FIGURE ACTION:

CREATE DIAGRAM.

FIGURE PURPOSE:

Show why two Planck likelihood constructions cannot be interpreted as the same objective with one additive extra penalty.

FIGURE TYPE:

Likelihood-architecture schematic.

FIGURE PROVENANCE:

Matched/full-multifrequency Planck bridge. B

Bubblevers 0.30m.docx

FIGURE PRODUCTION PROMPT:

TITLE OF IMAGE: The Planck Likelihood Construction Changes the Tested Geometry

TYPE: Scientific architecture diagram.

CONTENT: Create two parallel Planck analysis paths.

Path A: Matched CamSpec-lite n=3 EDE matched TT+TE+EE matched multipole support fixed- minus-free objective increase: +5.8238567748

Path B: Native full-multifrequency CamSpec n=3 EDE full frequency structure foreground and nuisance parameterization native covariance/likelihood architecture fixed-minus-free objective increase: +17.3979348723

Between the paths place a clearly marked warning: Different likelihood objectives Difference = diagnostic of likelihood-architecture dependence NOT an additive physical $\Delta\chi^2$ contribution

Show subsidiary components associated with the full-MF construction: frequency representation covariance foreground parameterization calibration/nuisance coupling

Do not draw a causal arrow from any individual component to the total penalty.

ASPECT RATIO: 16:9.

VISUAL STYLE: Technical publication diagram.

FIGURE CAPTION COPY BOX:

Figure 6. The matched CamSpec-lite and native full-multifrequency CamSpec calculations evaluate different Planck likelihood constructions. Their fixed-minus-free objective increases therefore diagnose likelihood-architecture dependence; their numerical difference is not an additional physical $\Delta\chi^2$ contribution.

Appendix D supplement — Q042 closure and Q043 repository preparation

Q042 is CLOSED with an inconclusive general-feasibility result. Its original scientific contract remains a 20-cell comparison of two native Planck likelihood arms, Λ CDM and $n=3$ EDE, ACT primary and lensing, DESI DR2, one identical supernova likelihood, and FULL plus four leave-one-out combinations. Contract SHA-256:

41e4d8f026933f64dbe41048f894002f479ba2b504432b33dc155f4dfe71378c.

Q042 reference truth is BLOCKED, its final result gate UNRESOLVED, and numerical production NOT AUTHORIZED. V29 completed finite initialized-context prerequisites only. V30/V31 yielded partial local conditional histories, not complete histories or a computed tau. The conditional cap argument found representation candidate counts of UPPER 138 and LOWER 3,002 against frozen cap32, subject to its explicitly stated admissibility, source/domain, continuation and unchanged-interval assumptions. These counts are not physical minima or a universal impossibility result.

Q043 (R-Q043-REPOSITORY-INTEGRATION-001) reached PASS_LOCAL_PREPARATION_AND_VALIDATION_ONLY. The proposed model candidate v0.4 / R000004 remains CANDIDATE_PREPARED_NOT_PROMOTED; accepted remote v0.3 / R000003 still covers Q001–Q041. The prepared package retains all 91 late-Q042 source objects, both claim-map trees, historical records, and the open portability prediction. Its proposed execution changes retire stale V29 ACTIVE eligibility without changing the permanent launcher or 43 unrelated registry entries.

Q043 internal validation included 8/8 candidate gates, 10/10 baseline/projected scientific checks, 13/13 formal checks, 10/10 historical Q041 checks, 10/10 failure-injection regressions, 20/20 public healthchecks, and 4/4 launcher resolver checks. These are technical consistency tests, not independent cosmological validation. Remote sync, admission, promotion and release were NOT PERFORMED.

Last inspected heads (historical inspection, not a new verification): model 10fa83797b2f7740519d4e43b1bebf0a6e35bada; execution 72cf9e92fc794c122f593a77b6a555e6e97f6a2e. The live registry still listed V29 ACTIVE. Controlled application and separately gated admission/promotion remain outstanding. No new numerical campaign is authorized.

Primary internal evidence: EVD-Q043-JOURNAL; EVD-Q043-RESULT; I-Q043-PREPARATION-VALIDATION-001; I-Q042-V31-INGESTION-001. Source register: q043_integration_result.json (SHA-256

4486e5173ba8ef3ac659c743bb578284c93446ce9e5b3d0dae71728beadbb9ff). The accepted Q041 release does not certify Q042.

Technical supplement | Q044

Scope of the mathematical result

Q044 is CLOSED with an inconclusive / insufficient-scientific-qualification outcome for the original matched-data CamSpec–HiLLiPoP H0/EDE question. The accepted local result R-Q044-SPLINE-ADJOINT-001 bounds a support-specific real-arithmetic functional under inherited source rails, fixed knots and stated source witnesses. It does not certify uniform native-binary rounding, a complete physical reionization history, or a reference-to-likelihood error envelope.

UPPER: 138 supports, indices 3195–3332; conditional native-plus enclosure [0.70676532823050475, 0.7067869701940579], width 2.1641963553142851e-05. LOWER: 3002 supports, indices 331–3332; enclosure [0.0016297839913017342, 0.0018091545475532051], width 0.00017937055625147098. These are dimensionless component bounds, not measured Thomson optical depths. Physically correlated source boxes can make independent-box extrema unattainable.

The received evidence reports 64 rational fixtures, 18 native-C diagnostics and 17 recovered V29 members; these tests were not independently rerun for this publication update. Finite diagnostics do not constitute a uniform arithmetic theorem.

Qualification and inference firewall

The frozen CLASS_EDE optical-depth integration uses a historical plus curvature correction whereas the ordinary cubic integral uses minus. The functional difference alone does not establish the physical bias of the frozen executable. The necessary dependency chain is complete-history qualification, physical reference accuracy, CMB TT/TE/EE accuracy, native likelihood transport, then converged posterior/model-decision qualification.

Q041 V19 produced no documented complete converged chains (0/32); Q042 did not qualify the original 20-cell inference matrix. Technical non-convergence is not physical falsification. Q043 subsequent repository installation establishes accepted v0.5/R000005 through Q043; Q044 is not remotely admitted. No production restart is authorized.

Reproducibility identifiers and next controlled test

Model repository: Morfindien/bubbleverse-model, inspected pin dd7575e7146382957d206e87274ca6246c2587b7. Execution repository: Morfindien/Bubbleverse, pin 72cf9e92fc794c122f593a77b6a555e6e97f6a2e. Native CLASS_EDE pin 5a131c91d657dd9a7c6364cc45b038710f8d0d97. Installation run 37927147545 is a technical installation receipt, not independent cosmological evidence.

Q045 remains PROPOSED, pending canonicalization: replace only the native optical-depth integration functional with a convergent Thomson-integral reference at identical requested tau_reio and unchanged physical inputs; quantify impacts on TT/TE/EE and native likelihoods relative to the frozen Q041 decision margins. Do not substitute a general software upgrade or an unqualified posterior campaign.

Internal source chain: I-Q044-SPLINE-ADJOINT-001; I-Q044-SPLINE-CHECKS-001; I-Q044-SPLINE-INGESTION-002; I-Q044-CONTRACT-001; I-Q044-M14-REPOSITORY-001. Contextual primary literature: arXiv:1605.03507v3; arXiv:1104.2932v2; arXiv:1104.2933.

Technical supplement · Q045 qualification and Q046 forward-history target

Status: Q045 CLOSED / PHYSICAL RESULT INCONCLUSIVE. Proposed Q046 not yet registered.

Q045 technical classification

Q045 is CLOSED with an inconclusive physical-materiality result: R-Q045-REFHIST-INGESTION-002. Remote V2 recovery completed successfully (GitHub Actions run 38042718814), yielding stored-history diagnostics without new theory calls. The original V1 native histories remain provenance, including their postprocessing failure. Technical success is not an independent physical-reference certificate.

Why the existing checks are insufficient

Three native refinement levels simultaneously vary nine numerical controls; reionization calibration and support selection also drift. All four accepted-tau sequences reverse direction. Same-history spline/quadrature agreement does not bound the error of the complete electron history. No complete-history remainder enclosure, isolated 10/01/11 intervention response, or original qualified inference margins are available. Starting-point likelihood gaps are not optimized decision margins.

Implementation and dependency chain

In the pinned CLASS_EDE source (commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97), the scalar reionization-calibration and cumulative-opacity paths use a plus curvature correction, while the ordinary cubic integral uses minus. This is a source-level functional distinction, not a demonstrated sign or magnitude of physical bias. Required qualification proceeds from complete physical history through Thomson depth and visibility, spectra, native likelihoods, and converged inference.

Q046 preregistration target

Proposed Q046, pending canonicalization: for four recovered Q041 LCDM and $n_{\text{scf}}=3$ EDE configurations, hold each L0 accepted z_{reio} fixed and compare the frozen electron fraction and cumulative Thomson depth against an independently integrated, complete event-resolved solution of the identical physical model. Freeze physical inputs, species and source definitions, background, events, endpoints, and support. Account separately for solver, interpolation, event, source, and integration error; preregister a justified comparison envelope before scientific interpretation.

Provenance

Execution repository: Morfindien/Bubbleverse at a9a5777b5923f32930cd9c84a49227ba36249326; V2 run 38042718814, job 114186014990. Model repository: Morfindien/bubbleverse-model at bba3528250b3d56c451bf5ca363b0d75120acb03. Reported accepted state: v0.6 / R000006 through Q044. These are supplied inspection records, not a fresh repository audit. Source: Q045 Universal Revision, 10 October 2026.